# PET CARE & VETERINARY APPOINTMENT MANAGEMENT SYSTEM

**A PROJECT REPORT**

*Submitted by*

**ASIFA H**

(241501025)

*in partial fulfilment of the End Semester Examination for the course*

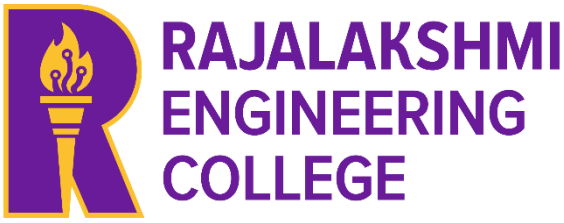
**CB23F35 –**

**LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR**

*in*

**BACHELOR OF TECHNOLOGY**

**ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING**




**RAJALAKSHMI ENGINEERING COLLEGE**

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

**OCTOBER 2026**

**RAJALAKSHMI ENGINEERING COLLEGE**

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# BONAFIDE CERTIFICATE

Certified that this project report titled "PET CARE & VETERINARY APPOINTMENT MANAGEMENT SYSTEM" is the bonafide work of ASIFA H (241501025), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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# DECLARATION

I, ASIFA H (241501025), hereby declare that the project report titled “PET CARE & VETERINARY APPOINTMENT MANAGEMENT SYSTEM”, submitted for the End Semester Examination of the course CB23F35 - Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application design described in this report - including its data model, forms, lookup relationships, reports, dashboard page, and proposed Deluge workflows - was prepared by me for this academic mini-project.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date:

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# ACKNOWLEDGEMENT

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**ASIFA H**

# ABSTRACT

Veterinary clinics routinely manage multiple interconnected administrative records, including pet owner contact details, individual pet profiles, veterinarian specializations and availability schedules, appointment bookings, and post-consultation medical histories. When these records are maintained manually in paper registers or disconnected spreadsheets, retrieving an owner's or pet's history becomes slow, appointment scheduling conflicts may arise, and tracking longitudinal diagnoses, treatments, and prescriptions requires repetitive administrative effort.

The Pet Care & Veterinary Appointment Management System is an academic mini-project designed using the Zoho Creator low-code development platform to centralize and streamline veterinary clinic administration. The system architecture is organized into five core relational forms—Pet Owners, Veterinarians, Pets, Appointments, and Medical Records—interconnected via lookup fields so that each pet is linked to its owner, each appointment references both a pet and a veterinarian, and each medical consultation record is associated with a specific appointment. Searchable reports (All Pets, Appointment Schedule, Appointment History, and Medical Records) and a centralized Dashboard page provide clinic staff with structured visibility into total registered pets, daily and upcoming appointments, and status breakdowns (Pending, Confirmed, Completed, and Cancelled). Configurable Zoho Creator workflows and illustrative Deluge scripts are designed to initialize appointment statuses, validate booking dates, check for veterinarian scheduling conflicts, and support optional email notifications.

Twelve structured test scenarios (TC01 to TC12) have been defined to verify form submissions, lookup integrity, mandatory field validation, date and conflict validation rules, status transitions, medical record creation, and dashboard summaries; in accordance with academic verification standards, execution results are marked as pending until tested in the live workspace. This report presents the complete low-code system design, implementation procedure, and governance framework, demonstrating how Zoho Creator simplifies veterinary clinic record management without requiring conventional full-stack programming.

***Keywords: Low-Code Development; Zoho Creator; Veterinary Appointment Management; Pet Care System; Lookup Relationships; Deluge Scripting; Workflow Automation***

# PROJECT AT A GLANCE

| Attribute | Details |
|---|---|
| **Project Title** | PET CARE & VETERINARY APPOINTMENT MANAGEMENT SYSTEM |
| **Application Domain** | Veterinary Clinic Administration & Pet Care Management |
| **Student Name & Register No.** | ASIFA H - 241501025 |
| **Zoho Creator Edition** | Zoho Creator (Account-dependent Education / Free / Trial Plan) |
| **Application Name (Link Name)** | Pet Care & Veterinary Appointment Management System |
| **Published / Live URL** | Academic Mini-Project Workspace (Not claimed as a public commercial deployment) |
| **Demo Credentials (per role)** | Clinic Administrator / Staff (Authenticated Zoho Creator Account) |
| **Number of Forms / Subforms** | 5 Core Forms (Pet Owners, Veterinarians, Pets, Appointments, Medical Records) / 0 Subforms |
| **Number of Reports (by type)** | List Reports: 4 (All Pets, Appointment Schedule, Appointment History, Medical Records) \| Calendar / Chart Views: Supported where configured |
| **Number of Pages / Dashboards** | 1 Centralized Clinic Dashboard Page (Proposed / Configurable) |
| **Number of Workflows** | 4 Designed Form Workflows (Workflow A: Status Init, Workflow B: Date Validation, Workflow C: Conflict Check, Workflow D: Confirmation Email) |
| **Deluge Functions / Custom Actions** | Illustrative Deluge Scripts for status initialization, date validation, conflict checking, and email notification |
| **Integrations** | Native Zoho Creator Email Notification action (Documented as proposed / subject to configuration); no external APIs |
| **Roles & Profiles Defined** | Clinic Staff / Receptionist, Veterinarian, Application Administrator (Configurable via Zoho Creator Permissions) |
| **Environments Used** | Zoho Creator Development Workspace |
| **Date of Production Deployment** | Academic Mini-Project Specification (October 2026) |

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# LIST OF ABBREVIATIONS

| Abbreviation | Expansion |
|---|---|
| **API** | Application Programming Interface |
| **CRUD** | Create, Read, Update, Delete |
| **DOB** | Date of Birth |
| **ER** | Entity–Relationship |
| **FK** | Foreign Key (Implemented via Lookup Field) |
| **FR** | Functional Requirement |
| **GPS** | Global Positioning System |
| **ID** | Identifier (Auto Number) |
| **LCAP** | Low-Code Application Platform |
| **NFR** | Non-Functional Requirement |
| **PK** | Primary Key |
| **RBAC** | Role-Based Access Control |
| **UI** | User Interface |
| **UAT** | User Acceptance Testing |

# CHAPTER 1
# INTRODUCTION

## 1.1 Background

Veterinary clinics and pet care centres manage a high volume of daily administrative and clinical records. Clinic receptionists and veterinarians must register pet owners, maintain detailed profiles for every animal (including species, breed, age, weight, allergies, and vaccination status), track veterinarian specializations and working days, schedule appointments, and record clinical consultations containing diagnoses, treatments, prescriptions, and follow-up dates.

In many small and medium veterinary practices, these operations rely on paper registers, appointment diaries, or disconnected spreadsheet files. Because owner contact records, pet profiles, doctor schedules, and consultation notes are stored separately, clinic staff must repeatedly re-enter owner and pet names, manually cross-check appointment slots, and search through physical folders to review a pet's prior medical history.

The Pet Care & Veterinary Appointment Management System has been designed as an academic mini-project for the course CB23F35 – Low-Code Application Development using Zoho Creator. It provides a centralized low-code application that connects pet owners, pets, veterinarians, appointments, and medical records through Zoho Creator forms, lookup relationships, searchable reports, a summary dashboard, and configurable Deluge workflows. This is an academic mini-project and is not represented as a certified veterinary medical system or a deployed commercial clinic platform.

All sample records referenced in this report—such as pet owners, pet names, veterinarian profiles, appointment bookings, and consultation notes—are strictly fictional demonstration data created for academic illustration.

## 1.2 Problem Statement

Veterinary clinics must coordinate multiple interrelated categories of information, including pet owner contact details, pet profiles, doctor availability, appointment schedules, and medical consultation histories. When this information is managed manually or across disconnected spreadsheets, retrieving accurate records becomes

slow, appointment scheduling conflicts may occur, consultation histories become difficult to trace, and repetitive data entry consumes valuable staff time. The specific administrative pain points addressed by this project are:

- Fragmented clinic records: Pet owner contacts, pet profiles, veterinarian details, appointment logs, and consultation notes are stored in separate registers or files.
- Repetitive manual data entry: Staff must repeatedly type owner names, phone numbers, and pet details whenever a new appointment or consultation is recorded.
- Appointment scheduling conflicts: Without centralized schedule views and validation rules, overlapping bookings for the same veterinarian and time slot can occur.
- Difficulty tracking pet medical histories: Retrieving a pet's past diagnoses, treatments, prescriptions, allergies, and follow-up dates across multiple visits is cumbersome.
- Lack of real-time operational visibility: Clinic staff lack a consolidated dashboard summarizing total registered pets, today's appointments, pending confirmations, and completed consultations.

### 1.3 Objectives

- To develop a centralized pet care and veterinary appointment management application on Zoho Creator and maintain organized records of pet owners and their contact details.
- To register individual pets (including species, breed, weight, allergies, and vaccination status) and associate each pet with its owner using lookup fields.
- To maintain veterinarian profiles, clinical specializations (General Veterinary Care, Dermatology, Surgery, Dentistry), available days, and consultation fees.
- To schedule and track veterinary appointments across Pending, Confirmed, Completed, and Cancelled statuses while linking each booking to a registered pet and veterinarian.
- To maintain structured medical consultation records—including diagnosis, treatment, prescription, follow-up date, and veterinarian notes—associated with appointments.

- To create searchable reports (All Pets, Appointment Schedule, Appointment History, Medical Records) and a centralized Dashboard displaying key appointment statistics.
- To apply data validation and configurable Zoho Creator / Deluge workflows (status initialization, date validation, conflict checking, and notifications) demonstrating low-code development in a veterinary clinic scenario.

### 1.4 Scope of the Project

The project covers core administrative and appointment management operations within a veterinary clinic using Zoho Creator. The in-scope functional capabilities include:

- Pet owner registration and contact information management (Owner ID, Name, Phone, Email, Address, Registration Date).
- Pet profile registration linked to owners and veterinarian profile management with specializations and available days.
- Veterinary appointment scheduling, status tracking (Pending, Confirmed, Completed, Cancelled), and medical consultation record maintenance.
- Searchable and filterable Zoho Creator reports (All Pets, Appointment Schedule, Appointment History, Medical Records) and a centralized Dashboard page.
- Workflow-based data validation, appointment conflict checking, status initialization, and email notifications where configured.

The application is designed to be accessed through a modern web browser on the Zoho Creator platform by authorized clinic staff and veterinarians.

**Out of scope. The project does not include online payment processing, pharmacy or medicine inventory management, live GPS pet tracking, AI-based autonomous veterinary diagnosis, telemedicine or video consultations, external laboratory integration, a public-facing customer self-service portal, or mobile push notifications. These items are discussed strictly as possible future enhancements in Chapter 9.**

### 1.5 Justification for a Low-Code Approach

A low-code application platform (LCAP) enables developers to construct data-driven applications through visual form builders, declarative lookup relationships, pre-built report views, and event-driven scripting rather than writing extensive full-stack

boilerplate code [1]. Zoho Creator provides an integrated cloud environment combining drag-and-drop form creation, relational lookup fields, tabular and calendar reports, a visual Page builder, and the Deluge scripting language [2].

The Pet Care & Veterinary Appointment Management System is ideally suited to Zoho Creator. Its core entities—Pet Owners, Veterinarians, Pets, Appointments, and Medical Records—map directly onto structured Zoho Creator forms connected via one-to-many lookup fields. Building an equivalent system using conventional frameworks (such as React, Java, Python, or Node.js with an external SQL database) would require separate front-end interfaces, REST API controllers, database schema migrations, and server hosting.

Using Zoho Creator significantly reduces repetitive coding, eliminates manual server provisioning, and allows forms, dropdown choices, reports, and validation workflows to be modified rapidly. At the same time, low-code development still requires rigorous engineering design: relational lookup integrity, mandatory field validation, appointment conflict prevention, and access permissions must be carefully planned and tested by the developer.

### 1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

# CHAPTER 2
# LITERATURE SURVEY AND SYSTEM STUDY

## 2.1 Study of the Existing System

In a traditional veterinary clinic without a centralized software system, administrative staff record pet owner details and pet descriptions in paper registers or standalone spreadsheets. When an owner calls or visits to book an appointment, the receptionist manually checks the veterinarian's working days in a separate diary, writes down the appointment date and time, and later hands a paper sheet to the veterinarian during the consultation. Diagnoses, treatments, prescriptions, and follow-up dates are written on physical cards or separate files. Figure 2.1 illustrates this traditional manual workflow.

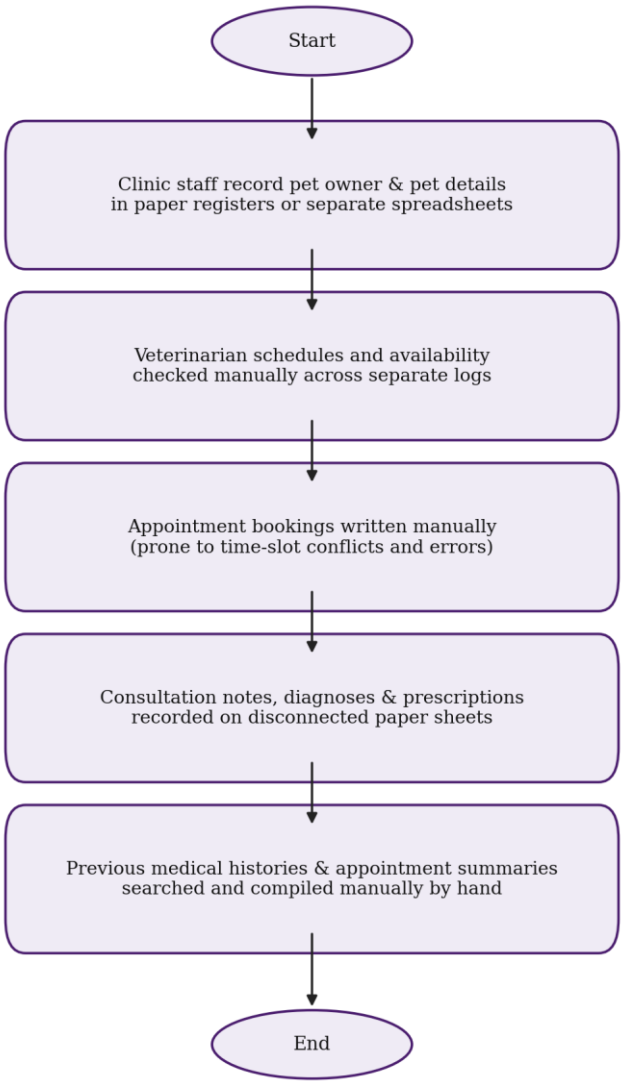



**Figure 2.1: As-Is Process Flow of the Existing (Manual) Veterinary Clinic System**

## 2.2 Limitations of the Existing System

- **Disconnected records: Owner contacts, pet profiles, doctor schedules, appointments, and medical records are stored in separate files or registers.**
- **Repetitive manual entry: Staff must re-write owner and pet details for every new appointment and consultation.**
- **Risk of scheduling conflicts: Double-booking a veterinarian for the same date and time slot can occur when schedules are checked manually.**
- **Slow retrieval of medical history: Reviewing a pet's past diagnoses, prescriptions, allergies, and vaccination status requires searching through multiple records.**
- **No consolidated clinic overview: Counting daily appointments, pending bookings, or completed consultations requires manual tallying.**

## 2.3 Low-Code versus Traditional Development

Table 2.1 compares a conventional full-stack (pro-code) web implementation of the Pet Care & Veterinary Appointment Management System with its low-code implementation on Zoho Creator, adapted from established low-code platform evaluation criteria [3].

**Table 2.1:** Comparison of Low-Code and Traditional Development for the Proposed Application

| Criterion | Traditional (Pro-Code) Development | Low-Code on Zoho Creator |
|---|---|---|
| **Development effort** | Longer: requires separate relational database setup, backend API coding, and frontend UI forms. | Shorter: forms, lookup fields, reports, and dashboard pages are configured visually in Zoho Creator. |
| **Skills required** | Full-stack web development (e.g., HTML/JS/React, Java/Python, SQL, server deployment). | Relational data modelling, Zoho Creator configuration, and Deluge scripting for custom workflows. |
| **Database & relationships** | Manual SQL table creation, primary/foreign key constraints, and JOIN queries. | Zoho Creator forms act as structured record collections linked via Auto Number IDs and Lookup fields. |
| **Hosting & infrastructure** | Web server and database server must be provisioned, secured, and maintained separately. | Cloud-hosted directly on the Zoho Creator platform; no separate server setup is required. |

| Criterion | Traditional (Pro-Code) Development | Low-Code on Zoho Creator |
|---|---|---|
| **Workflow & notifications** | Custom event listeners and SMTP mail server integration must be programmed. | Built-in Zoho Creator Form Workflows, Deluge triggers, and configurable email notification actions. |
| **Security & access control** | Custom authentication, session management, and role middleware must be coded. | Uses Zoho account authentication and configurable platform permission sets. |
| **Maintenance & modification** | Adding new fields or changing choices requires code edits, testing, and redeployment. | Fields, dropdown choices, and reports can be updated directly through the visual builder. |
| **Customisation & plan limits** | Complete control over custom UI widgets and third-party libraries. | Bounded by Zoho Creator visual components and the capabilities/quotas of the selected account plan. |

## 2.4 Literature Survey

The literature and technical documentation reviewed for this project comprise academic studies on low-code development platforms [1], [3], [4] and official Zoho Creator developer documentation covering forms, lookup fields, reports, Pages, workflows, Deluge scripting, and permissions [2], [5]–[8]. Table 2.2 summarizes these sources and their relevance to the veterinary clinic management system.

**Table 2.2:** Summary of Literature Survey

| S.No. | Author(s) / Source & Year | Title / Product | Key Findings | Relevance to this Project |
|---|---|---|---|---|
| 1 | A. C. Bock and U. Frank, 2021 [1] | Low-Code Platform (Business & Information Systems Engineering) | Analyses low-code platforms as model-driven environments that reduce hand-written code for structured business applications. | Justifies selecting a low-code platform for managing structured veterinary clinic entities and workflows. |
| 2 | Zoho Corporation [2] | Zoho Creator Help Documentation – Application & Form Builder | Provides official specifications for creating Zoho Creator applications, forms, Auto Number fields, and choice fields. | Primary reference for designing the Pet Owners, Veterinarians, Pets, Appointments, and Medical Records forms. |

| S.No. | Author(s) / Source & Year | Title / Product | Key Findings | Relevance to this Project |
|---|---|---|---|---|
| 3 | A. Sahay et al., 2020 [3] | Supporting the Understanding and Comparison of Low-Code Development Platforms | Establishes a taxonomy for comparing data modeling, user interface design, and workflow automation in low-code platforms. | Provides the evaluation framework used in Table 2.1 to compare low-code and traditional development. |
| 4 | D. Di Ruscio et al., 2022 [4] | Low-Code Development and Model-Driven Engineering | Explains how declarative form and workflow models generate underlying storage, validation, and user interfaces. | Supports the 4-layer platform architecture (UI, Logic, Data Management, Automation) documented in Chapter 4. |
| 5 | Zoho Corporation [5], [6] | Zoho Creator Lookup Fields, Reports, and Pages Documentation | Details how Lookup fields establish one-to-many relationships and how Reports and Pages aggregate and display records. | Guides the relational links (Owner->Pet->Appointment->Medical Record) and clinic dashboard design. |
| 6 | Zoho Corporation [7], [8] | Zoho Creator Workflows, Deluge Scripting, and Permissions Guide | Documents form-event triggers (On Validate, On Success), Deluge syntax, email tasks, and role permissions. | Informs Workflows A–D (status initialization, date validation, conflict check, and confirmation alerts). |

## 2.5 Emerging Trends in Low-Code Development

**Visual and model-driven application development. Modern low-code platforms allow developers to define data structures, field validations, and relational lookups visually while automatically generating the underlying storage and tabular reports [1], [4]. In this project, the five veterinary clinic forms act as the core structural models of the application.**

**Event-driven workflow automation. Rather than writing monolithic backend controllers, low-code platforms attach concise scripts to specific form lifecycle events (such as record creation, pre-submission validation, or status updates) [7], [8]. In Zoho Creator, Deluge workflows enable targeted rules such as initializing appointment status to Pending or checking for scheduling conflicts.**

**Centralized operational dashboards. Low-code platforms provide drag-and-drop page builders that combine summary KPI cards, filtered report views, and status charts on a single screen [6], enabling clinic staff to monitor daily appointments and pending requests without manual tallying.**

**Responsible academic scoping of low-code systems. While enterprise low-code platforms can integrate with external APIs or AI services, an academic mini-project must clearly distinguish between core form/report configuration, proposed Deluge workflow automation, and out-of-scope commercial features (such as AI diagnosis or telemedicine).**

### 2.6 Proposed System

The proposed Pet Care & Veterinary Appointment Management System is a centralized Zoho Creator application organized around five interconnected forms—Pet Owners, Veterinarians, Pets, Appointments, and Medical Records—together with four searchable reports (All Pets, Appointment Schedule, Appointment History, and Medical Records), a centralized Clinic Dashboard page, and configurable Deluge workflows.

**Record Linkage: Pet Owners (1:N) → Pets (1:N) → Appointments ← (N:1) Veterinarians; Appointments (1:1 / 1:N) → Medical Records**

Each entity uses a dedicated Auto Number field (Owner_ID, Doctor_ID, Pet_ID, Appointment_ID, Record_ID) as its unique identifier rather than relying on display names, while Lookup fields link related records across forms. Medical diagnoses, treatments, and prescriptions are entered strictly by authorized veterinary personnel; the system does not perform autonomous medical diagnosis.

The proposed design directly resolves the limitations of manual clinic management: disconnected registers are unified into one relational Zoho Creator workspace; repetitive typing is eliminated through Lookup dropdowns; appointment tracking is structured across Pending, Confirmed, Completed, and Cancelled statuses; and consultation histories are linked directly to each appointment and pet.

# CHAPTER 3
# REQUIREMENT ANALYSIS

## 3.1 Stakeholders and User Roles

**Table 3.1:** Stakeholders and User Roles

| Role | Description | Key Responsibilities in the Application |
|---|---|---|
| **Clinic Staff / Receptionist** | Administrative personnel managing front-desk clinic operations. | Register pet owners and pets, maintain veterinarian details, schedule appointments, update booking statuses, and search reports. |
| **Veterinarian** | Veterinary doctors providing clinical consultations and treatments. | Review assigned appointments, inspect pet profiles and allergies, and enter medical records (diagnosis, treatment, prescription, follow-up date). |
| **Application Developer / Administrator** | Student developer (ASHVANTH M – 241501024) building and configuring the Zoho Creator application. | Create forms, configure lookup relationships and choice fields, build reports and the dashboard page, and define Deluge workflows and permissions. |
| **Pet Owners (Indirect Stakeholders)** | Clients who bring their pets to the veterinary clinic for care. | Do not log in directly (no public portal in current scope); their contact details and pets are managed by clinic staff, and they may receive email confirmations if configured. |

## 3.2 Functional Requirements

**Table 3.2:** Functional Requirements

| FR ID | Requirement | Role(s) | Priority |
|---|---|---|---|
| FR-01 | The system shall maintain Pet Owner records including Owner ID (Auto Number), Owner Name, Phone, Email, Address, and Registration Date. | Clinic Staff | High |
| FR-02 | The system shall maintain Veterinarian profiles including Doctor ID, Doctor Name, Specialization, Phone, Email, Available Days, and Consultation Fee. | Clinic Staff, Admin | High |
| FR-03 | The system shall register Pets (Pet ID, Pet Name, Species, Breed, DOB, Gender, Weight, Allergies, Vaccination Status) and associate each pet with an existing Pet Owner via a Lookup field. | Clinic Staff | High |
| FR-04 | The system shall schedule Appointments (Appointment ID, Pet Lookup, Veterinarian Lookup, Date, Time, Reason, Status, Notes) across Pending, Confirmed, Completed, and Cancelled statuses. | Clinic Staff, Veterinarian | High |

| FR ID | Requirement | Role(s) | Priority |
|---|---|---|---|
| FR-05 | The system shall support workflow initialization of new Appointments to 'Pending' status (Workflow A) and validate that Appointment Date is not in the past (Workflow B, where configured). | System (Workflow / Deluge) | High |
| FR-06 | The system design shall support checking existing Appointments to prevent conflicting bookings for the same Veterinarian, Date, and Time slot (Workflow C, where implemented and tested). | System (Workflow / Deluge) | Medium |
| FR-07 | The system shall allow authorized veterinary personnel to record Medical Records (Record ID, Appointment Lookup, Diagnosis, Treatment, Prescription, Follow-Up Date, Veterinarian Notes). | Veterinarian | High |
| FR-08 | The system shall provide searchable and filterable reports: All Pets Report, Appointment Schedule Report, Appointment History Report, and Medical Records Report. | Clinic Staff, Veterinarian | High |
| FR-09 | The system shall provide a centralized Dashboard Page displaying summary metrics (Total Pets, Today's, Pending, Completed, and Upcoming Appointments) and an Appointment Status Chart. | Clinic Staff, Veterinarian | Medium |
| FR-10 | The system shall support sending an email notification to the pet owner when an Appointment status changes to 'Confirmed' (Workflow D, if email notification is configured). | System (Workflow) | Low (Optional) |

## 3.3 Non-Functional Requirements

**Table 3.3:** Non-Functional Requirements

| NFR ID | Category | Requirement |
|---|---|---|
| NFR-01 | Usability & Clarity | Forms, lookup dropdowns, and reports shall use clear field labels and logical grouping so clinic staff can enter and retrieve records efficiently. |
| NFR-02 | Relational Integrity | Auto Number fields shall serve as unique identifiers, and Lookup fields shall ensure pets, appointments, and medical records reference valid parent records. |
| NFR-03 | Data Validation | Mandatory fields, email/phone format checks, and date/conflict validation rules shall reduce incomplete or invalid clinic records. |
| NFR-04 | Security & Privacy | Access to pet owner contact details and veterinary medical records shall be restricted to authenticated Zoho Creator users according to configured permissions. |
| NFR-05 | Platform Compatibility | The application shall be accessible through current versions of standard web browsers (Chrome, Edge, Firefox, Safari) over a stable internet connection. |
| NFR-06 | Maintainability | Forms, choice lists, reports, and workflows shall be modularly structured in Zoho Creator to allow easy future enhancements. |

## 3.4 Hardware and Software Requirements

**Table 3.4:** Hardware and Software Requirements

| Category | Specification |
|---|---|
| **Hardware – Device** | Desktop computer or laptop with keyboard and pointing device |
| **Hardware – Resources** | Sufficient memory (RAM) and processing capability to run a modern web browser |
| **Network Connectivity** | Stable internet connection (required for accessing cloud-based Zoho Creator) |
| **Operating System** | Microsoft Windows, macOS, Linux, or another supported desktop operating system |
| **Web Browser** | Current version of a supported browser (Google Chrome, Microsoft Edge, Mozilla Firefox, or Safari) |
| **Development Platform** | Zoho Creator (Low-Code Application Development Platform) |
| **Account & Plan** | Authenticated Zoho account (Availability of specific workflow/email quotas depends on the Zoho Creator plan) |
| **Scripting Environment** | Zoho Creator built-in Deluge Script Editor |

### 3.5 Use Case Model

Figure 3.1 illustrates the use case model of the Pet Care & Veterinary Appointment Management System. Clinic Staff and Veterinarians act as the primary operational actors managing owners, pets, veterinarians, appointments, medical records, reports, and the dashboard, while the Application Developer/Administrator configures the forms, lookups, and workflows. Table 3.5 details the core appointment scheduling and consultation use case.

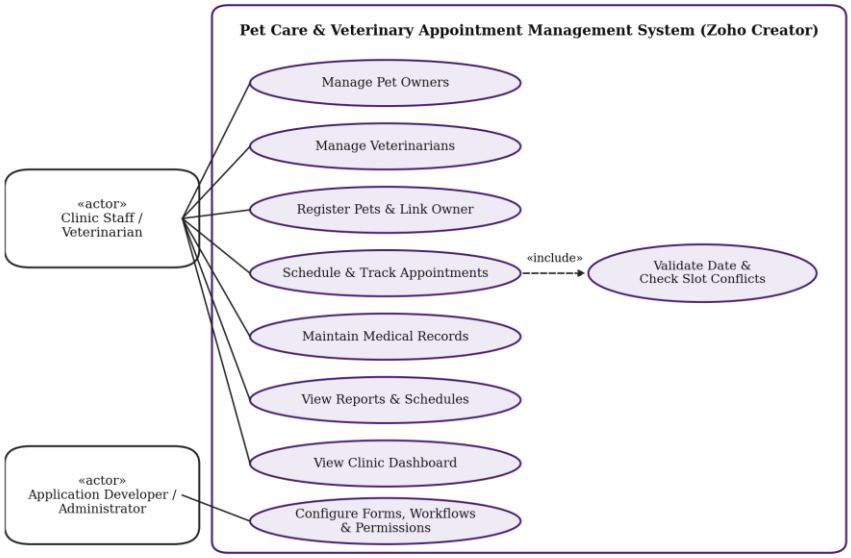



**Figure 3.1: Use Case Diagram of the Pet Care & Veterinary Appointment Management System**

**Table 3.5: Use Case Description – Schedule Veterinary Appointment and Record Consultation**

| | |
|---|---|
| **Use Case ID** | UC-01 |
| **Use Case Name** | Schedule Veterinary Appointment and Record Medical Consultation |
| **Actor(s)** | Clinic Staff / Receptionist, Veterinarian |
| **Pre-condition** | Pet Owner, Pet profile (linked to Owner), and Veterinarian profile are registered in Zoho Creator. |
| **Main Flow** | 1. Clinic staff opens the Appointments form and selects the Pet and Veterinarian via Lookup fields. 2. Staff selects the Appointment Date, Appointment Time, and Reason, then submits the form. 3. Configured validation rules verify the date and check for scheduling conflicts; status is initialized to Pending. 4. Upon confirmation and visit completion, status is updated to Confirmed and Completed. 5. Veterinarian opens the Medical Records form, selects the Appointment, and records Diagnosis, Treatment, Prescription, Follow-Up Date, and Notes. |

| | |
|---|---|
| **Alternate Flow** | If a mandatory field is missing, the date is in the past (Workflow B), or the selected Veterinarian already has an active appointment in that Date/Time slot (Workflow C, if configured), submission is blocked with a validation message. |
| **Post-condition** | Validated appointment and medical consultation records are stored and reflected in the Appointment Schedule, Medical Records Report, and Clinic Dashboard. |
| **Related FR** | FR-04, FR-05, FR-06, FR-07, FR-08, FR-09 |

# CHAPTER 4
# SYSTEM DESIGN

## 4.1 System Architecture

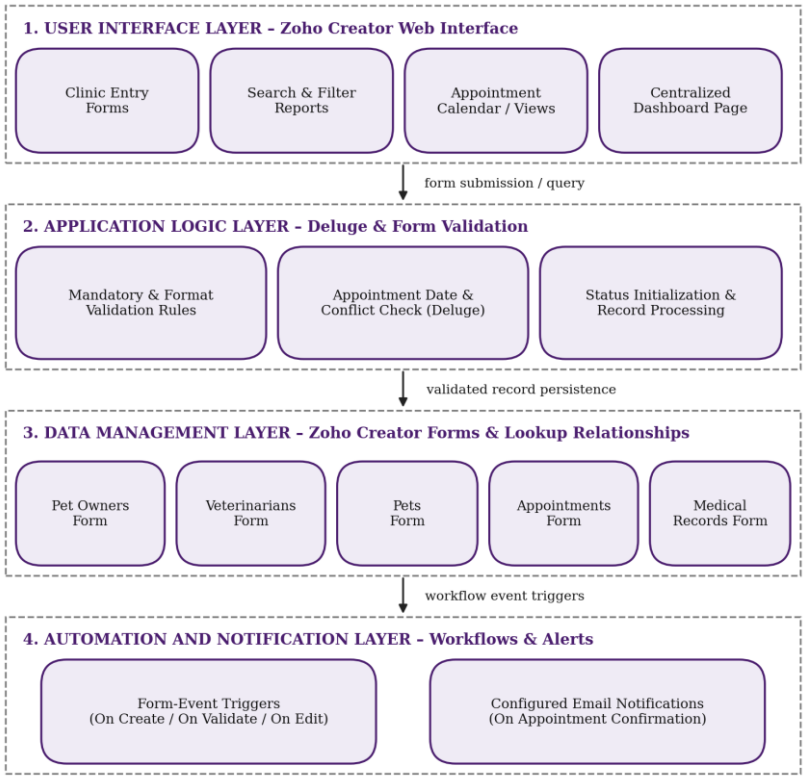



**Figure 4.1: System Architecture of the Pet Care & Veterinary Appointment Management System**

As shown in Figure 4.1, the Pet Care & Veterinary Appointment Management System is structured into four logical layers within Zoho Creator (no independent MySQL server, React frontend, Node.js backend, or external ML model is used):

**1. User Interface Layer: Comprises the visual Zoho Creator data-entry forms (Pet Owners, Veterinarians, Pets, Appointments, Medical Records), searchable tabular/calendar reports, and the centralized Dashboard page accessed via a web browser.**

**2. Application Logic Layer: Encompasses mandatory field rules, choice constraints, workflow conditions, and Deluge scripts responsible for status initialization, appointment date validation, and veterinarian conflict checking.**

**3. Data Management Layer: Consists of structured record collections stored natively within Zoho Creator forms and interconnected through Auto Number identifiers and Lookup relationships.**

**4. Automation and Notification Layer: Executes event-driven workflow actions upon form creation, validation, or status modification, including configured email notifications when appointments are confirmed.**

## 4.2 Data Model

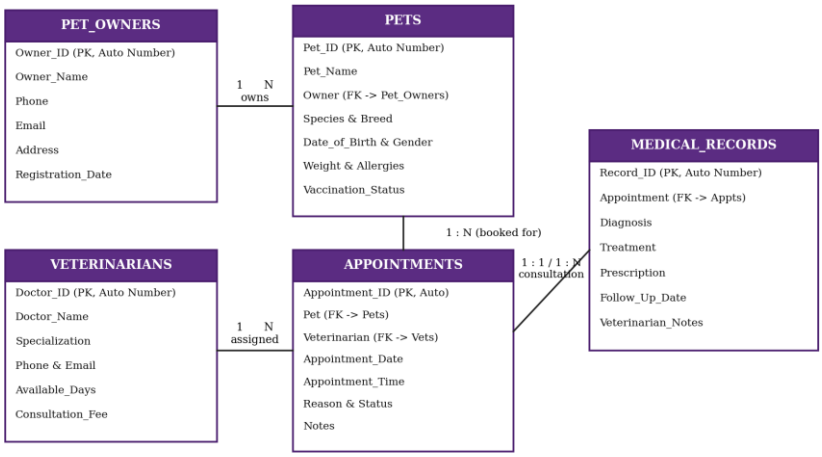



**Figure 4.2: Entity–Relationship Diagram of the Pet Care & Veterinary Appointment Management System**

In Zoho Creator, each form functions as a structured database table (record collection). Dedicated Auto Number fields (Owner_ID, Doctor_ID, Pet_ID, Appointment_ID, Record_ID) provide unique record identifiers rather than relying on owner or pet display names, while Lookup fields establish foreign-key relationships across the five entities. Note that storing Available Days in Veterinarians does not automatically guarantee real-time slot availability unless conflict-checking logic is implemented, and the relationship between Appointments and Medical Records is 1:1 conceptually (or 1:N if multiple consultation entries per appointment are permitted).

**Table 4.1:** Entities and Relationships

| Entity (Form) | Key Fields | Related To | Relationship Type | Implemented Using |
|---|---|---|---|---|
| Pet Owners | Owner_ID (Auto Number), Owner_Name, Phone, Email, Address, Registration_Date | Pets | One-to-Many (1:N) | Lookup field (Owner) in the Pets form referencing Pet Owners |
| Veterinarians | Doctor_ID (Auto Number), Doctor_Name, Specialization, Phone, Email, Available_Days, Consultation_Fee | Appointments | One-to-Many (1:N) | Lookup field (Veterinarian) in the Appointments form referencing Veterinarians |
| Pets | Pet_ID (Auto Number), Pet_Name, Owner (Lookup), Species, Breed, Date_of_Birth, Gender, Weight, Allergies, Vaccination_Status | Pet Owners (N:1); Appointments (1:N) | One-to-Many (1:N to Appointments) | Lookup field (Pet) in the Appointments form referencing Pets |
| Appointments | Appointment_ID (Auto Number), Pet (Lookup), Veterinarian (Lookup), Appointment_Date, Appointment_Time, Reason, Status, Notes | Pets, Veterinarians (N:1); Medical Records (1:1 / 1:N) | One-to-One / One-to-Many to Medical Records | Lookup field (Appointment) in the Medical Records form referencing Appointments |
| Medical Records | Record_ID (Auto Number), Appointment (Lookup), Diagnosis, Treatment, Prescription, Follow_Up_Date, Veterinarian_Notes | Appointments | Many-to-One / One-to-One | Lookup field (Appointment) linking clinical notes to the specific booking |

### 4.3 Form Design

Tables 4.2 to 4.5 specify the complete field definitions, Deluge link names, Zoho Creator field types, mandatory constraints, and choice values for the five application forms: Pet Owners, Veterinarians, Pets, Appointments, and Medical Records.

**Table 4.2: Field Specification – Module 1: Pet Owners Form**

| Field Label | Link Name | Field Type | Mandatory | Validation / Default | Remarks |
|---|---|---|---|---|---|
| Owner ID | Owner_ID | Auto Number | — | System-generated unique ID (e.g., OWN-001) | Primary identifier for pet owner |
| Owner Name | Owner_Name | Name / Single Line | Yes | Required text entry | Full name of the pet owner |
| Phone | Phone | Phone | Yes | Valid phone number format | Primary contact number |
| Email | Email | Email | Yes | Valid email format | Used for appointment confirmation alerts |
| Address | Address | Address / Multi Line | No | Residential address text | Owner residential contact address |
| Registration Date | Registration_Date | Date | Yes | Default: Current Date (zoho.currentdate) | Date the owner was registered |

**Table 4.3: Field Specification – Module 2: Veterinarians Form**

| Field Label | Link Name | Field Type | Mandatory | Validation / Default | Remarks |
|---|---|---|---|---|---|
| Doctor ID | Doctor_ID | Auto Number | — | System-generated unique ID (e.g., DOC-001) | Primary identifier for veterinarian |
| Doctor Name | Doctor_Name | Name / Single Line | Yes | Required text entry | Full name of the veterinarian |
| Specialization | Specialization | Drop Down | Yes | Choices: General Veterinary Care, Dermatology, Surgery, Dentistry | Clinical specialization area |
| Phone | Phone | Phone | Yes | Valid phone format | Veterinarian contact phone |
| Email | Email | Email | Yes | Valid email format | Veterinarian official email |
| Available Days | Available_Days | Multi Select | Yes | Choices: Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday | Indicates general working days (does not alone prevent slot conflicts) |
| Consultation Fee | Consultation_Fee | Currency | Yes | Positive currency value (INR) | Standard consultation charge |

**Table 4.4: Field Specification – Module 3: Pets Form**

| Field Label | Link Name | Field Type | Mandatory | Validation / Default | Remarks |
|---|---|---|---|---|---|
| Pet ID | Pet_ID | Auto Number | — | System-generated unique ID (e.g., PET-001) | Unique identifier for each pet |
| Pet Name | Pet_Name | Single Line | Yes | Required text entry | Display name of the pet |
| Owner | Owner | Lookup (Pet Owners) | Yes | Must reference an existing Pet Owner record | Establishes 1:N Owner-to-Pets relationship |
| Species | Species | Drop Down | Yes | Choices: Dog, Cat, Bird, Rabbit, Other | Animal species category |
| Breed | Breed | Single Line | No | Free-text breed name (e.g., Labrador, Persian) | Specific breed of the pet |
| Date of Birth | Date_of_Birth | Date | No | Must not be a future date | Used to determine pet age |
| Gender | Gender | Drop Down | Yes | Choices: Male, Female, Unknown | Biological sex of the pet |
| Weight | Weight | Decimal | No | Positive decimal value (in kg) | Recorded weight for clinical reference |
| Allergies | Allergies | Multi Line | No | Free-text allergy notes (or 'None known') | Known drug or food allergies |
| Vaccination Status | Vaccination_Status | Drop Down | Yes | Choices: Up to Date, Due Soon, Overdue, Not Vaccinated | Current immunization status |

**Table 4.5: Field Specification – Module 4: Appointments Form & Module 5: Medical Records Form**

| Field Label | Link Name | Field Type | Mandatory | Validation / Default | Remarks |
|---|---|---|---|---|---|
| Appointments: Appointment ID | Appointment_ID | Auto Number | — | System-generated unique ID (e.g., APT-001) | Unique identifier for appointment |
| Appointments: Pet | Pet | Lookup (Pets) | Yes | References registered Pet record | Links booking to pet (and its owner) |

| Field Label | Link Name | Field Type | Mandatory | Validation / Default | Remarks |
|---|---|---|---|---|---|
| Appointments: Veterinarian | Veterinarian | Lookup (Veterinarians) | Yes | References registered Veterinarian | Assigns doctor for consultation |
| Appointments: Appointment Date | Appointment_Date | Date | Yes | Validated via Deluge: Date >= Current Date | Scheduled date of visit |
| Appointments: Appointment Time | Appointment_Time | Drop Down / Time | Yes | Slot choices (e.g., 09:00 AM, 10:00 AM, 11:00 AM, 02:00 PM, 04:00 PM) | Used with Veterinarian + Date for conflict check |
| Appointments: Reason | Reason | Drop Down | Yes | Choices: General Checkup, Vaccination, Dermatology, Surgery, Dental Care, Emergency, Follow-Up | Primary purpose of visit |
| Appointments: Status | Status | Drop Down | Yes | Choices: Pending (Default), Confirmed, Completed, Cancelled | Tracks appointment lifecycle |
| Appointments: Notes | Notes | Multi Line | No | Optional pre-visit notes or symptoms | Additional booking remarks |
| Medical Records: Record ID | Record_ID | Auto Number | — | System-generated unique ID (e.g., MED-001) | Unique identifier for consultation record |
| Medical Records: Appointment | Appointment | Lookup (Appointments) | Yes | References existing Appointment record | Links clinical record to appointment |
| Medical Records: Diagnosis | Diagnosis | Multi Line | Yes | Entered by authorized veterinarian | Clinical findings and diagnosis |
| Medical Records: Treatment | Treatment | Multi Line | Yes | Entered by authorized veterinarian | Procedures or treatment administered |
| Medical Records: Prescription | Prescription | Multi Line | No | Medication names, dosage, and duration | Prescribed medications |
| Medical Records: Follow-Up Date | Follow_Up_Date | Date | No | Optional next review date (> Appointment_Date) | Recommended follow-up visit date |

| Field Label | Link Name | Field Type | Mandatory | Validation / Default | Remarks |
|---|---|---|---|---|---|
| Medical Records: Veterinarian Notes | Veterinarian_Notes | Multi Line | No | Dietary or recovery instructions | Additional clinical observations |

## 4.4 Process and Workflow Design

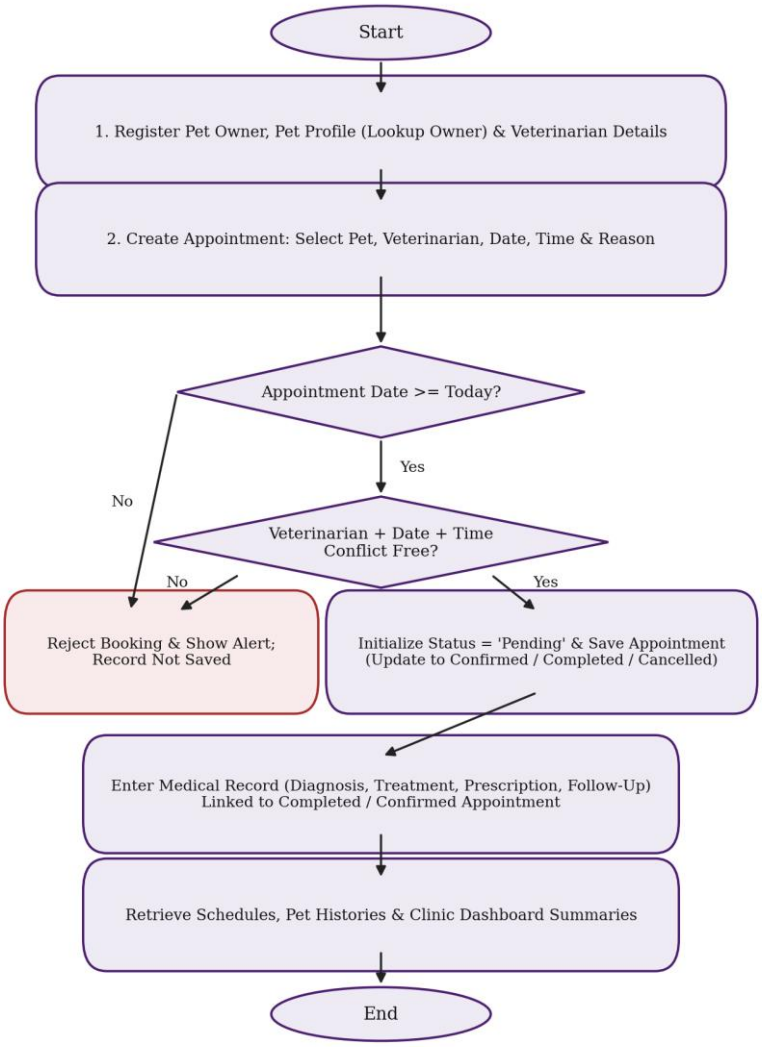
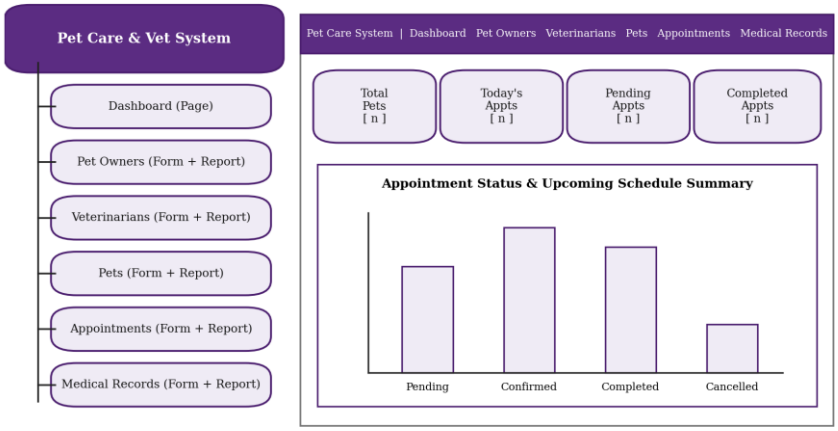



**Figure 4.3: Workflow / Process Flow for Appointment Scheduling, Validation, and Consultation Recording**

Figure 4.3 illustrates the ten-step operational workflow of the system: (1) Clinic staff register a pet owner; (2) The pet is registered and linked to its owner via Lookup; (3) Veterinarian profiles and specializations are maintained; (4) An appointment is created by selecting the pet, veterinarian, date, and time; (5) Appointment validation runs according to configured rules (checking that the date is not in the past and checking

whether the veterinarian already has an active booking in that time slot); (6) Valid bookings have their status initialized to Pending (while invalid bookings are rejected); (7) The appointment status is updated to Confirmed, Completed, or Cancelled as appropriate; (8) Authorized veterinary personnel enter a Medical Record (diagnosis, treatment, prescription, follow-up date) linked to the appointment; (9) Staff retrieve schedules and histories via reports; and (10) The Dashboard summarizes current clinic records.

### 4.5 User Interface and Navigation Design




*(a) Navigation structure* *(b) Dashboard wireframe (conceptual layout)*

**Figure 4.4: Navigation Structure and Wireframe of the Clinic Dashboard Page**

The application navigation is organized into six sections matching its functional modules: Dashboard, Pet Owners, Veterinarians, Pets, Appointments, and Medical Records (Figure 4.4a). Figure 4.4b presents the conceptual wireframe of the centralized Dashboard Page, featuring summary metric cards (Total Registered Pets, Today's Appointments, Pending Appointments, Completed Appointments, and Upcoming Appointments) above an Appointment Status Chart. All dashboard figures are derived dynamically from underlying form records and report aggregations rather than static numbers.

### 4.6 Access Control Design

Table 4.6 defines the intended Role–Permission Matrix using A = Add, E = Edit, V = View, D = Delete, and — = No access. Clinic Administrators have full control; Receptionist/Clinic Staff manage owners, pets, and appointment bookings; Veterinarians view schedules and enter/update clinical Medical Records; and Pet

Owners do not have direct portal login access in the current scope. Permissions must be explicitly configured in Zoho Creator settings.

**Table 4.6:** Role–Permission Matrix

| Application Component | Administrator (Owner) | Clinic Staff / Receptionist | Veterinarian | Pet Owner (Public) |
|---|---|---|---|---|
| Form: Pet Owners | A E V D | A E V | V | — |
| Form: Veterinarians | A E V D | V | E V (own profile) | — |
| Form: Pets | A E V D | A E V | E V | — |
| Form: Appointments | A E V D | A E V | E V | — |
| Form: Medical Records | A E V D | V | A E V | — |
| Reports & Clinic Dashboard | V | V | V | — |

# CHAPTER 5
# IMPLEMENTATION

## 5.1 Application Setup

The chronological development procedure in Zoho Creator follows ten structured steps: Step 1 (Create the Application): Sign in to Zoho Creator, select Create Solution / Create from Scratch, and name the application 'Pet Care & Veterinary Appointment Management System'. Step 2 (Create Forms): Use the drag-and-drop visual builder to create the Pet Owners, Veterinarians, Pets, Appointments, and Medical Records forms. Step 3 (Configure Lookup Fields): Link Pets.Owner → Pet Owners, Appointments.Pet → Pets, Appointments.Veterinarian → Veterinarians, and Medical_Records.Appointment → Appointments. Step 4 (Configure Choice Fields): Set up dropdown and multi-select options for Species, Gender, Specialization, Available Days, Vaccination Status, Appointment Reason, and Appointment Status. Steps 5 to 10 encompass configuring Reports, building the Dashboard Page, adding Workflows/Deluge scripts, entering fictional sample data, testing, and reviewing sharing settings.

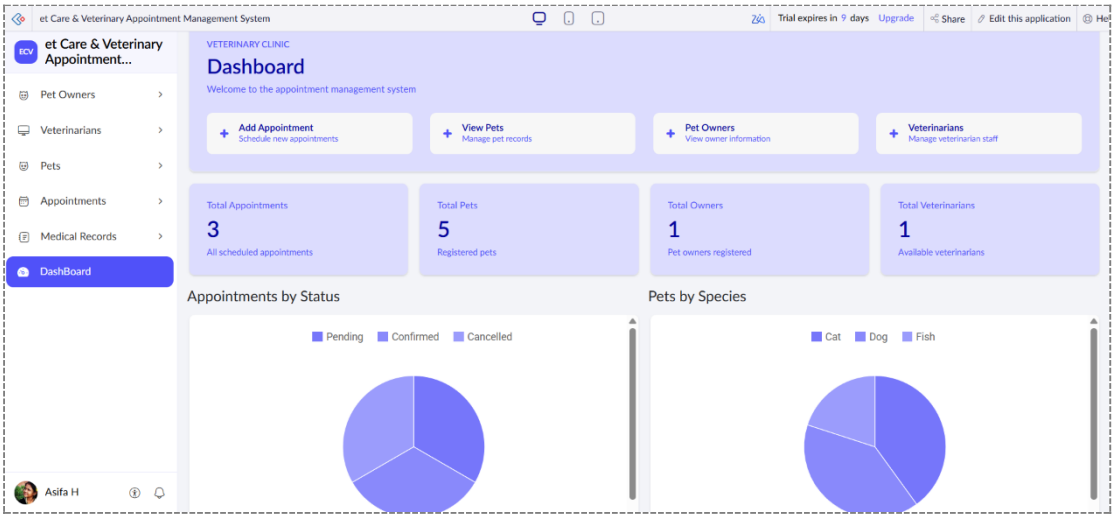



**Figure 5.1: Application Home and Navigation Sections (Placeholder)**

## 5.2 Forms and Data Capture

Five interconnected forms handle data collection and storage in Zoho Creator. Each form automatically provisions a structured record collection with an Auto Number primary identifier and built-in field validation:

**Module 1 – Pet Owners Form & Module 2 – Veterinarians Form: The Pet Owners form captures Owner_ID (Auto Number), Owner_Name, Phone, Email, Address, and Registration_Date, enabling clinic staff to register and retrieve client contact profiles. The Veterinarians form records Doctor_ID (Auto Number), Doctor_Name, Specialization (General Veterinary Care, Dermatology, Surgery, Dentistry), Phone, Email, Available_Days (Multi Select), and Consultation_Fee (Currency). Note that storing Available_Days provides scheduling reference information but does not automatically enforce real-time slot availability without additional Deluge validation.**

**Module 3 – Pet Registration Form: The Pets form stores Pet_ID (Auto Number), Pet_Name, Owner (Lookup to Pet Owners), Species (Dropdown), Breed, Date_of_Birth, Gender, Weight (Decimal), Allergies, and Vaccination_Status (Up to Date, Due Soon, Overdue, Not Vaccinated). Using a Lookup field to Pet Owners allows one owner to be associated with multiple pets without duplicating owner contact details.**

**Module 4 – Veterinary Appointment Management Form: The Appointments form captures Appointment_ID (Auto Number), Pet (Lookup to Pets), Veterinarian (Lookup to Veterinarians), Appointment_Date, Appointment_Time, Reason, Status (Pending, Confirmed, Completed, Cancelled), and Notes. Because a basic form alone does not prevent overlapping bookings, conflict prevention requires workflow validation against existing appointment records.**

**Module 5 – Medical Record Management Form: The Medical Records form captures Record_ID (Auto Number), Appointment (Lookup to Appointments), Diagnosis, Treatment, Prescription, Follow_Up_Date, and Veterinarian_Notes. All clinical diagnoses, treatments, and prescriptions are entered manually by authorized veterinary personnel; the application does not generate autonomous medical diagnoses.**

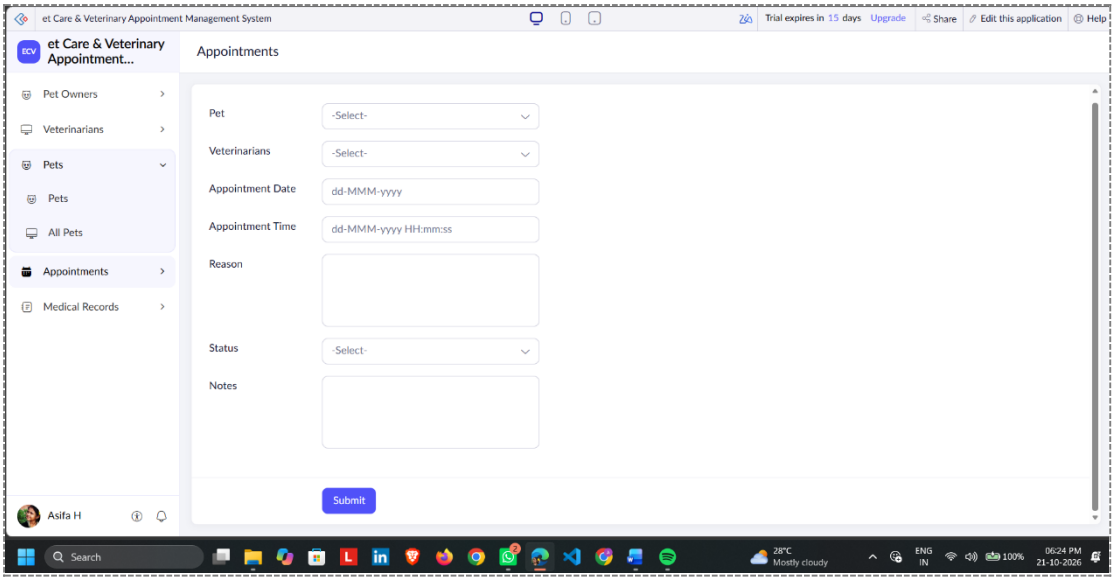



**Figure 5.2: Form Configuration in the Zoho Creator Form Builder (Placeholder)**

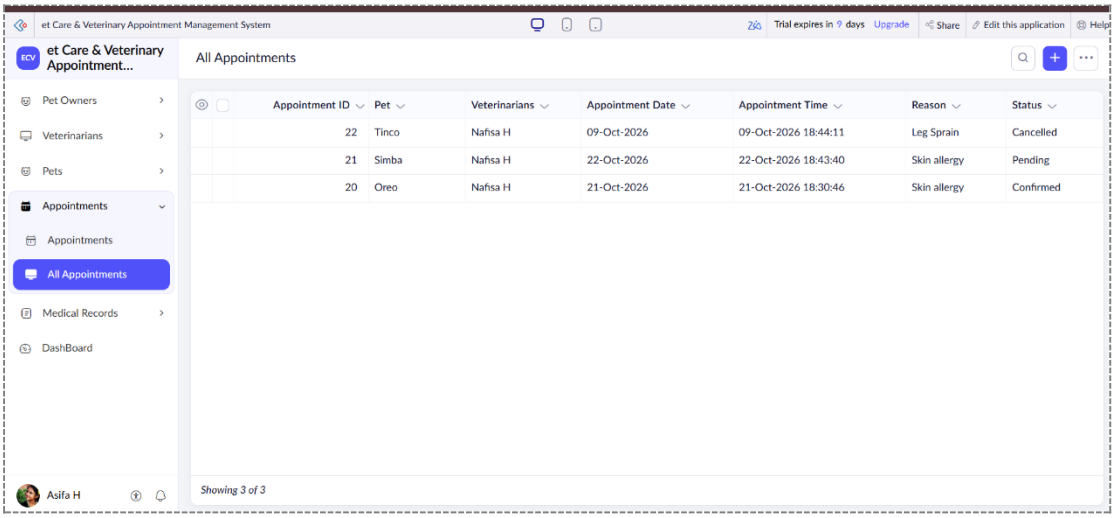



**Figure 5.3: Appointments Form in Live Mode (Placeholder)**

## 5.3 Reports

Module 6 – Reports and Search: Zoho Creator reports enable clinic staff and veterinarians to display, search, filter, sort, and review stored records across all five forms. Where supported by the account plan, a calendar-style report can also be configured on the Appointments form to visualize scheduled visits by date.

**Four core tabular reports are defined in Table 5.1: (1) the All Pets Report displaying pet names, species, breed, linked owner, and vaccination status; (2) the Appointment Schedule Report displaying appointment IDs, pets, veterinarians, dates, times, and current statuses; (3) the Appointment History Report allowing staff to filter completed, cancelled, pending, and confirmed appointments; and (4)**

**the Medical Records Report displaying consultation diagnoses, treatments, prescriptions, and follow-up dates linked to appointments.**

**Table 5.1:** Reports Implemented

| Report Name | Type | Source Form | Filter / Grouping / Sorting | Custom Actions | Roles |
|---|---|---|---|---|---|
| All Pets Report | List Report | Pets | Search by Pet_Name / Owner; Filter by Species & Vaccination_Status | View / Edit Pet Profile | Clinic Staff, Veterinarian |
| Appointment Schedule Report | List / Calendar Report | Appointments | Filter by Appointment_Date, Veterinarian, & Status; Sort by Date/Time | Update Status | Clinic Staff, Veterinarian |
| Appointment History Report | List Report | Appointments | Grouped/Filtered by Status (Completed, Cancelled, Pending, Confirmed) | View Consultation History | Clinic Staff, Veterinarian |
| Medical Records Report | List Report | Medical_Records | Search by Appointment / Pet; Filter by Follow_Up_Date | View / Edit Clinical Notes | Veterinarian, Admin |
| All Pet Owners Report | List Report | Pet_Owners | Search by Owner_Name, Phone, or Email | View Linked Pets | Clinic Staff, Admin |
| Veterinarians Directory Report | List Report | Veterinarians | Filter by Specialization & Available_Days | View Assigned Appointments | Clinic Staff, Admin |

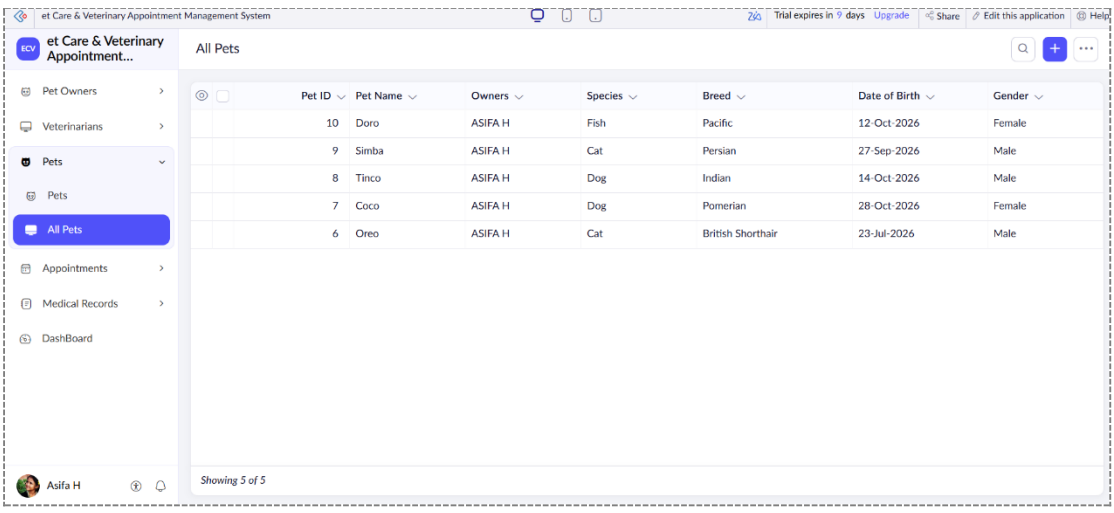


**Figure 5.4: Appointment Schedule and All Pets Report View (Placeholder)**

## 5.4 Pages and Dashboards

Module 7 – Dashboard: A centralized Dashboard Page can be constructed using Zoho Creator Pages by combining KPI summary panels, embedded report views, and chart components. The designed clinic dashboard includes summary cards for Total Registered Pets, Today's Appointments, Pending Appointments, Completed Appointments, and Upcoming Appointments, alongside an Appointment Status Chart grouping bookings by status. All dashboard metrics must be connected to actual form records and report aggregations rather than hardcoded numbers.

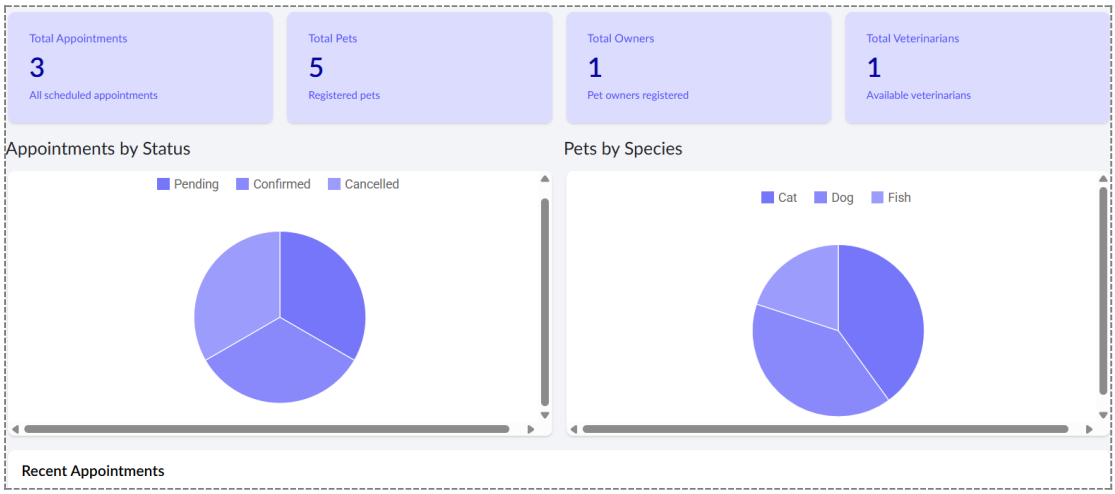



**Figure 5.5: Clinic Dashboard Page in Zoho Creator (Placeholder)**

## 5.5 Deluge Scripting

Deluge (Data Enriched Language for the Universal Grid Environment) is Zoho Creator's built-in scripting language used to implement conditional validation, default field initialization, conflict checks, record updates, and workflow notifications. Table 5.2 summarizes the four illustrative Deluge scripts designed for the Pet Care & Veterinary Appointment Management System. Because internal field link names depend on the exact form configuration in Zoho Creator, these scripts are presented as illustrative design logic whose field link names must match the live Form Builder.

**Table 5.2:** Deluge Scripts Implemented

| Script ID | Location / Trigger | Purpose | Deluge Concepts Used |
|---|---|---|---|
| DS-01 (Workflow A) | Appointments → Form Workflow (On Load / On Create) | Initialize new appointment records with Status = 'Pending' where configured. | input field assignment, string literal |

| Script ID | Location / Trigger | Purpose | Deluge Concepts Used |
|---|---|---|---|
| DS-02 (Workflow B) | Appointments → Form Workflow (On Validate) | Validate Appointment_Date and reject past dates (Appointment_Date < zoho.currentdate). | if condition, null check, zoho.currentdate, alert, cancel submit |
| DS-03 (Workflow C) | Appointments → Form Workflow (On Validate) | Check existing active appointments for the selected Veterinarian, Date, and Time to prevent double-booking. | collection fetch criteria, count(), alert, cancel submit |
| DS-04 (Workflow D) | Appointments → Field/Form Workflow (On Edit -> On Success) | Send an email notification to the pet owner when Appointment Status changes to 'Confirmed' (if configured). | lookup record fetch, sendmail task |

***Code 5.1: Illustrative Deluge Script for Appointment Date & Conflict Validation (Workflows B & C – On Validate)***

```
// Script ID: DS-02 & DS-03 | Form: Appointments | Event: Created
or Edited -> On Validate
// Illustrative Deluge script (verify exact internal field link
names in Zoho Creator)

// Workflow B: Reject missing or past appointment dates
if(input.Appointment_Date == null || input.Appointment_Date <
zoho.currentdate)
{
    alert "Invalid Date: Appointment Date cannot be in the past.";
    cancel submit;
}

// Workflow C: Prevent double-booking for the same Veterinarian,
Date, and Time
existing_appts = Appointments[Veterinarian == input.Veterinarian
&& Appointment_Date == input.Appointment_Date && Appointment_Time
== input.Appointment_Time && Status != "Cancelled" && ID !=
input.ID];
if(existing_appts.count() > 0)
{
    alert "Scheduling Conflict: The selected veterinarian already
has an active appointment at this date and time.";
    cancel submit;
}
```

***Code 5.2: Illustrative Deluge Script for Status Initialization (Workflow A) & Confirmation Email (Workflow D)***

```
// Workflow A (DS-01) | Form: Appointments | Event: On Load (New
Record)
if(input.Status == null || input.Status == "")
{
    input.Status = "Pending";
}

// Workflow D (DS-04) | Form: Appointments | Event: Edited -> On
Success (When Status == "Confirmed")
// Note: Requires verified sender/recipient email configuration in
Zoho Creator plan
if(input.Status == "Confirmed")
{
    pet_rec = Pets[ID == input.Pet];
    owner_rec = Pet_Owners[ID == pet_rec.Owner];
    if(owner_rec.Email != null && owner_rec.Email != "")
    {
        sendmail
        [
            from : zoho.adminuserid
            to : owner_rec.Email
            subject : "Veterinary Appointment Confirmed – " +
pet_rec.Pet_Name
            message : "Dear " + owner_rec.Owner_Name + ", your
appointment for " + pet_rec.Pet_Name + " on " +
input.Appointment_Date + " at " + input.Appointment_Time + " has
been confirmed."
        ];
    }
}
```

In Code 5.1, the On Validate trigger executes before a record is committed to the Zoho Creator database. Workflow B compares input.Appointment_Date against zoho.currentdate and cancels submission if a past date is selected. Workflow C queries the Appointments form for any non-cancelled booking matching the same Veterinarian, Appointment_Date, and Appointment_Time (excluding the current record's ID during edits) and blocks conflicting bookings.

In Code 5.2, Workflow A initializes new bookings to the 'Pending' status, while Workflow D illustrates how an On Success workflow fetches the linked Pet and Pet_Owners records by ID to send a confirmation email. In accordance with academic

accuracy standards, these scripts are documented as proposed/illustrative workflow logic to be verified against the live account configuration.

### 5.6 Workflows and Automation

Module 8 – Workflow Automation: Zoho Creator workflows consist of a trigger (form event), optional filter criteria, and one or more actions (Deluge scripts, field updates, or email notifications). Table 5.3 distinguishes the four workflows designed for the veterinary appointment system.

**Table 5.3:** Workflows Implemented

| Workflow Name | Type | Trigger / Event | Condition | Action(s) | Script Ref. |
|---|---|---|---|---|---|
| Workflow A: Appointment Registration | Form Workflow (Proposed / Configurable) | Appointments – On Load / On Create | New appointment record created | Initialize Status field to 'Pending' | DS-01 |
| Workflow B: Appointment Date Validation | Form Workflow (Proposed / Configurable) | Appointments – On Validate | Appointment_Date < zoho.currentdate | Display alert and cancel form submission | DS-02 |
| Workflow C: Appointment Conflict Validation | Form Workflow (Proposed / Configurable) | Appointments – On Validate | Matching Veterinarian + Date + Time with Status != 'Cancelled' | Display conflict alert and prevent double-booking | DS-03 |
| Workflow D: Appointment Confirmation Alert | Form Workflow (Proposed / Account-Dependent) | Appointments – On Edit (On Success) | Status changes to 'Confirmed' | Send confirmation email to pet owner via sendmail | DS-04 |

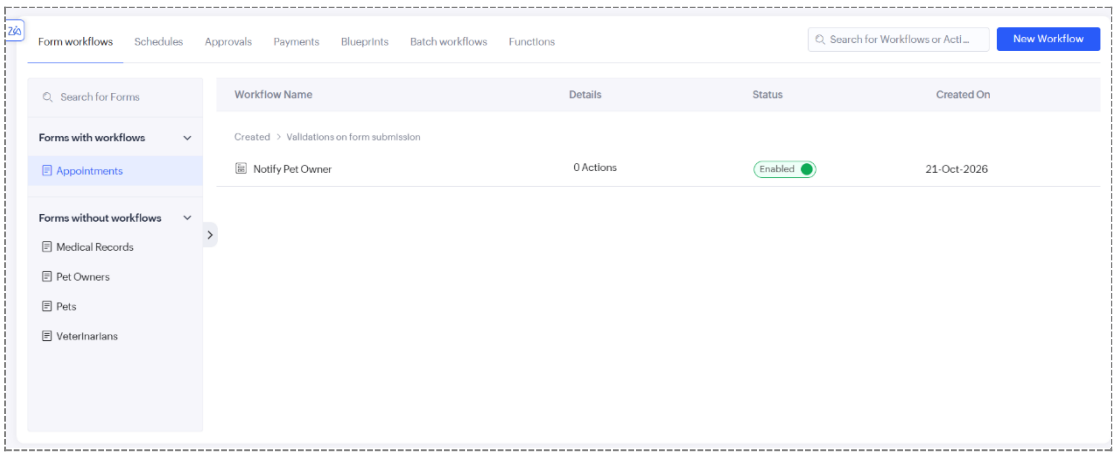



**Figure 5.6: Appointments Form Workflow Configuration (Placeholder)**

## 5.7 Integrations

No external third-party REST API, payment gateway, external laboratory system, or external database integration is used in this academic mini-project. All records are stored natively within Zoho Creator forms. Email notifications (Workflow D) rely on Zoho Creator's built-in sendmail workflow action, subject to account plan limits and verified email configuration.

**Table 5.4:** Integrations Implemented

| Integration | Service / Endpoint | Method | Authentication | Data Exchanged / Purpose |
|---|---|---|---|---|
| Workflow Email Notification | Zoho Creator Built-in Mail Action | Deluge sendmail / Email Task | Authenticated Zoho Creator Account | Send appointment confirmation updates to pet owners (Proposed / if configured) |
| External APIs / Payment Gateways | Not used | Not implemented | — | Out of scope for the academic mini-project |

***Code 5.3: External REST API Integration – Not Applicable to Current Academic Scope***

```
// Not applicable: The Pet Care & Veterinary Appointment
Management System
// uses native Zoho Creator forms, lookups, reports, and internal
workflows.
// No external REST API or third-party service integration is
claimed.
```

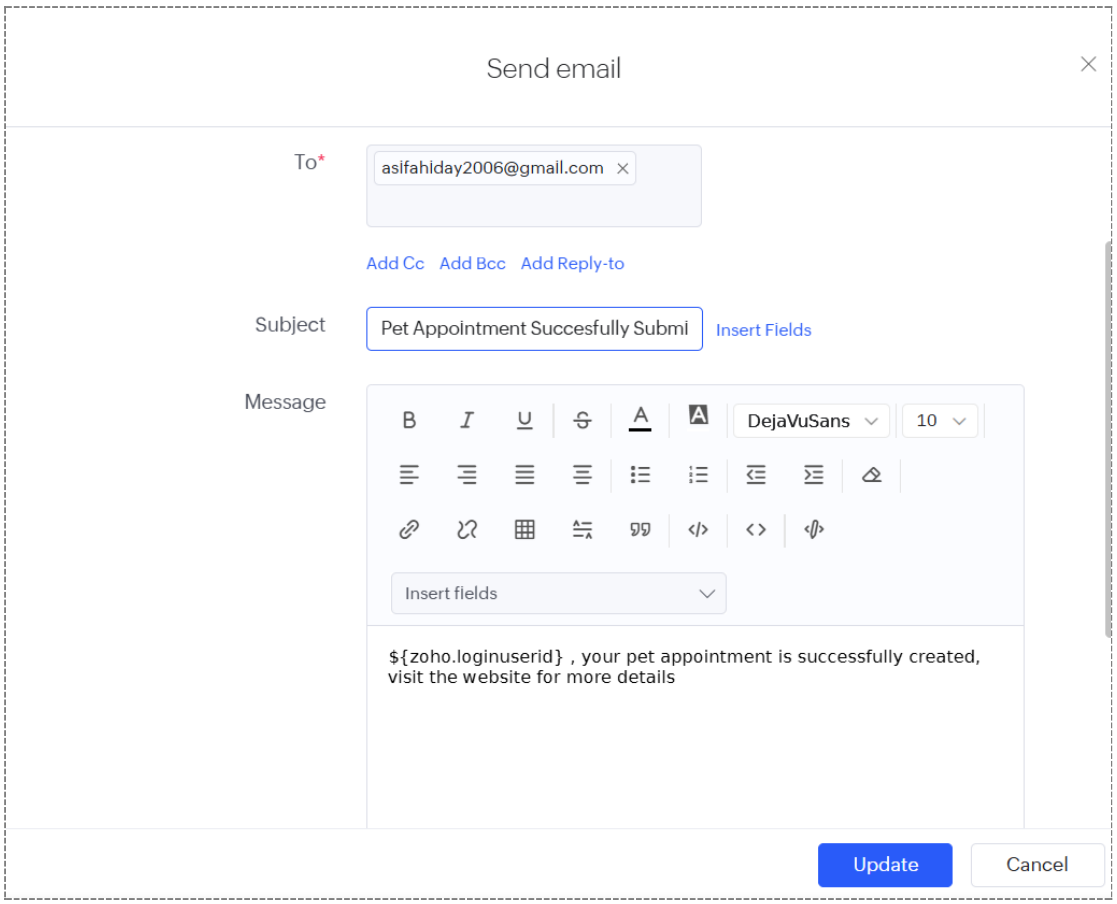



**Figure 5.7: Email Notification Action Configuration (Placeholder / If Configured)**

### 5.8 User Management and Role-Based Access

Access to the application is controlled through Zoho account authentication. Within Zoho Creator, role-based permissions can be configured to restrict access by user role (Clinic Administrator, Receptionist/Staff, and Veterinarian) as designed in Table 4.6, ensuring that pet owners' personal contact details and clinical medical records are protected from unauthorized viewing or modification.

In a basic student developer workspace, the application is operated primarily under the authenticated owner account. Specific custom roles or external sharing permissions should be documented as active only if explicitly configured and verified in the account settings.

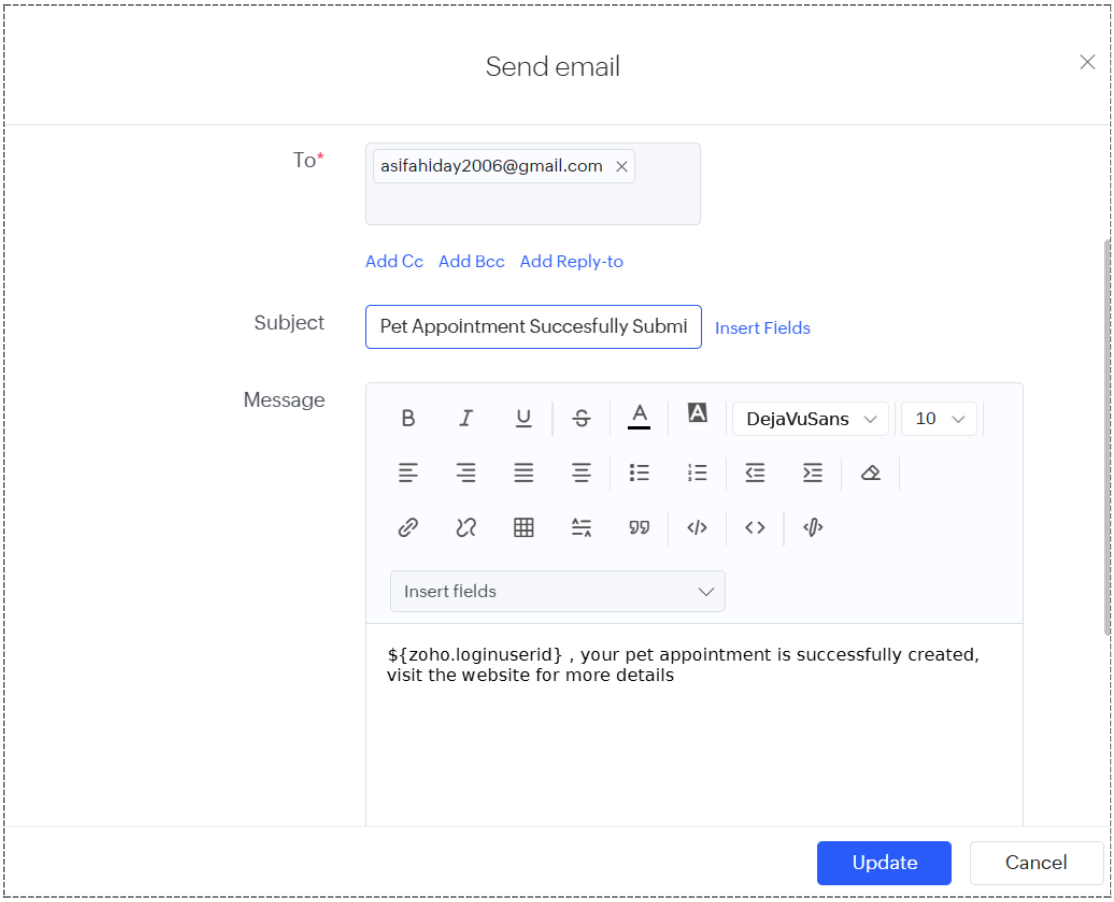



**Figure 5.8: Application Permissions and Access Settings (Placeholder)**

### 5.9 AI and Analytics Features

The Pet Care & Veterinary Appointment Management System does not use artificial intelligence or machine learning models to diagnose animal diseases or prescribe treatments. All clinical diagnoses, treatments, and prescriptions in the Medical Records form are entered by authorized veterinary personnel.

Analytics in the application consist of descriptive operational aggregations on the Dashboard Page and in Zoho Creator Reports—such as total registered pets, daily and upcoming appointment counts, and an Appointment Status Chart (Pending, Confirmed, Completed, Cancelled) derived from stored appointment records.

### 5.10 Pro-Code Extension

No pro-code extensions (such as Zoho Catalyst microservices, custom JavaScript widgets, or custom full-stack React/Node.js servers) are used in this mini-project. The administrative requirements of registering owners, pets, veterinarians, appointments,

and medical consultations are addressed through standard Zoho Creator forms, lookup relationships, reports, Pages, and Deluge scripts.

Pro-code extensions or external integrations could be considered in future iterations if the clinic required online payment gateway processing, automated pharmacy inventory deduction, or telemedicine video consultations.

### 5.11 Mobile and Portal Access

The application is designed for access via a standard desktop or laptop web browser by clinic staff and veterinarians. A public-facing customer self-booking portal and mobile push notifications are not included in the current project scope and are listed as future enhancements.

# CHAPTER 6
# TESTING AND DEBUGGING

## 6.1 Testing Strategy

The testing and validation plan for the Pet Care & Veterinary Appointment Management System verifies form submission, mandatory field enforcement, lookup integrity, date and conflict validation rules, status transitions, report filtering, and dashboard aggregation across twelve formal test scenarios (TC01 to TC12):

- **Form & Mandatory Field Validation (TC01, TC04): Verify that valid Pet Owner records save properly and that missing mandatory fields on the Appointments form block submission.**
- **Lookup Relationship Verification (TC02, TC03, TC08, TC12): Confirm that Pets link accurately to Pet Owners, Appointments link to Pets and Veterinarians, and Medical Records link to Appointments.**
- **Workflow & Deluge Validation Testing (TC05, TC06, TC07, TC11): Test past-date rejection (Workflow B), veterinarian time-slot conflict prevention (Workflow C), status updates from Pending to Confirmed, and email notification triggers (Workflow D, if configured).**
- **Report Filtering & Dashboard Verification (TC09, TC10): Verify that the Appointment Schedule / History reports filter accurately by status and that Dashboard summary cards match the actual stored records.**
- **Execution Status Disclosure: In strict compliance with academic verification standards, because live test execution evidence is not attached in this document, Actual Result is recorded as 'Not yet tested' and Status is marked as 'Pending' in Table 6.1.**

## 6.2 Test Cases

**Table 6.1:** Test Cases

| TC ID | FR Ref. | Test Description | Input / Steps | Expected Result | Actual Result | Status |
|---|---|---|---|---|---|---|
| TC01 | FR-01 | Submit a valid pet owner record | Open Pet Owners form; enter fictional Owner Name, valid Phone, valid Email, Address, and Registration Date; click Submit. | Owner record is saved with a unique Auto Number Owner_ID and appears in the Pet Owners report. | Not yet tested | Pending |
| TC02 | FR-03 | Register a pet and associate it with an existing owner | Open Pets form; enter Pet Name, select an existing Owner from the Lookup dropdown, select Species, Gender, and Vaccination Status; click Submit. | Pet record is saved with a unique Pet_ID and linked to the selected Pet Owner record. | Not yet tested | Pending |
| TC03 | FR-04 | Create an appointment with valid pet and veterinarian references | Open Appointments form; select an existing Pet and Veterinarian via Lookup fields, choose a valid future Appointment Date, Time, and Reason; submit. | Appointment record is created with a unique Appointment_ID and initialized to 'Pending' status. | Not yet tested | Pending |
| TC04 | FR-04, NFR-03 | Attempt to submit an appointment with a missing mandatory field | Open Appointments form; leave Pet, Veterinarian, or Appointment Date blank and attempt to submit. | Zoho Creator displays a mandatory field validation error and prevents record submission. | Not yet tested | Pending |
| TC05 | FR-05 | Test appointment date validation (Workflow B) | Open Appointments form; select a past date (prior to current date) for Appointment Date and submit. | Configured On Validate Deluge script (DS-02) displays an invalid date alert and cancels submission. | Not yet tested | Pending |

| TC ID | FR Ref. | Test Description | Input / Steps | Expected Result | Actual Result | Status |
|---|---|---|---|---|---|---|
| TC06 | FR-06 | Test appointment conflict prevention (Workflow C) | Attempt to submit a second appointment for the same Veterinarian, Appointment Date, and Appointment Time as an existing non-cancelled booking. | Configured On Validate conflict script (DS-03) detects the duplicate slot, displays a conflict alert, and blocks submission. | Not yet tested | Pending |
| TC07 | FR-04 | Update an appointment from Pending to Confirmed | Open an existing Pending appointment in the Appointment Schedule Report, edit Status to 'Confirmed', and save. | Appointment status updates to 'Confirmed' and reflects immediately in the schedule report and dashboard. | Not yet tested | Pending |
| TC08 | FR-07 | Create a medical record linked to an appointment | Open Medical Records form; select an existing Appointment via Lookup, enter Diagnosis, Treatment, Prescription, and Follow-Up Date; submit. | Medical record is saved with a unique Record_ID and linked to the selected Appointment. | Not yet tested | Pending |
| TC09 | FR-08 | Filter appointments by status | Open Appointment Schedule / History Report and apply filter Status == 'Pending' (or 'Completed'). | Report displays only appointment records matching the selected status filter. | Not yet tested | Pending |
| TC10 | FR-09 | Verify dashboard summaries against stored records | Open the Clinic Dashboard Page and compare Total Pets, Pending Appointments, and Completed Appointments cards against report counts. | Dashboard summary counts and status chart match the exact number of records stored in the underlying forms. | Not yet tested | Pending |

| TC ID | FR Ref. | Test Description | Input / Steps | Expected Result | Actual Result | Status |
|---|---|---|---|---|---|---|
| TC11 | FR-10 | Test email notification if implemented (Workflow D) | Update an appointment Status to 'Confirmed' when Workflow D email notification is enabled. | If configured, confirmation email is sent to the pet owner's registered email address. | Not yet tested | Pending |
| TC12 | FR-03, FR-04, FR-07 | Verify that lookup fields reference the correct records | Inspect Lookup dropdowns in Pets (Owner), Appointments (Pet, Veterinarian), and Medical Records (Appointment). | Each Lookup field lists and links to the exact corresponding parent records via their internal record IDs. | Not yet tested | Pending |

All twelve test cases (TC01 to TC12) are marked as 'Pending' ('Not yet tested') until executed against the live Zoho Creator application with verified screenshots.

### 6.3 Debugging Deluge Scripts

When configuring Deluge scripts in Zoho Creator, scripts can be validated and debugged using: (i) the Deluge editor's built-in syntax checker to verify internal field link names; (ii) 'info' statements to inspect lookup IDs and criteria query counts during test runs; (iii) 'alert' boxes paired with 'cancel submit' on On Validate events; and (iv) testing boundary conditions such as past dates and overlapping time slots.

Table 6.2 provides the defect log structure for recording any configuration or scripting issues encountered during live testing (such as mismatched internal field link names or self-matching records during edit-mode conflict checks, which is prevented by including ID != input.ID in the criteria query).

### 6.4 Defect Log

**Table 6.2:** Defect Log

| Bug ID | Description | Root Cause | Fix Applied | Status |
|---|---|---|---|---|
| BUG-01 (Illustrative) | Edit-mode conflict check could flag the record being edited as a duplicate slot | Criteria query checked Veterinarian, Date, and Time without excluding the current record ID | Include && ID != input.ID in the DS-03 criteria query | Design Verified |

| Bug ID | Description | Root Cause | Fix Applied | Status |
|---|---|---|---|---|
| BUG-02 (Pending Live Test) | To be recorded during live form and workflow execution | — | — | Pending |

### 6.5 User Acceptance Testing

User acceptance testing (UAT) is to be conducted after full form and workflow configuration by having clinic staff/student reviewers test owner registration, pet lookup selection, appointment scheduling, and consultation entry. Table 6.3 provides the UAT recording template.

**Table 6.3:** User Acceptance Feedback

| User | Role Tested | Feedback | Action Taken |
|---|---|---|---|
| Reviewer 1 (Pending) | Clinic Staff / Receptionist Role | Not yet tested (To be recorded during live demonstration) | Pending |
| Reviewer 2 (Pending) | Veterinarian Role | Not yet tested (To be recorded during live demonstration) | Pending |

# CHAPTER 7
# DEPLOYMENT, SECURITY AND GOVERNANCE

## 7.1 Environments and Deployment Strategy

In Zoho Creator, applications are built and previewed inside the developer's workspace and can be shared with specific authorized users according to the capabilities of the selected Zoho Creator plan. Step 10 of the development procedure involves reviewing all field properties, mandatory constraints, lookup relationships, workflows, and reports before accessing the live mode.

Because this project is an academic mini-project, it is developed and demonstrated within the Zoho Creator workspace and is not claimed as a publicly deployed commercial clinic platform or external customer portal.

**Figure 7.1: Application Preview and Sharing Settings (Placeholder)**

**Table 7.1:** Deployment Checklist

| S.No. | Checklist Item | Done (✓) |
|---|---|---|
| 1 | Create Pet Owners, Veterinarians, Pets, Appointments, and Medical Records forms | Designed / To Verify |
| 2 | Configure Lookup fields (Owner, Pet, Veterinarian, Appointment) and Choice fields | Designed / To Verify |
| 3 | Configure All Pets, Appointment Schedule, Appointment History, and Medical Records reports | Designed / To Verify |
| 4 | Build centralized Clinic Dashboard Page with summary cards and status chart | Designed / To Verify |
| 5 | Configure Deluge workflows (Status initialization, Date validation, Conflict check) | Proposed / To Verify |
| 6 | Enter fictional sample data and execute test cases TC01 to TC12 | Pending Execution |
| 7 | Review access permissions to protect owner contact details and medical records | Pending Verification |

## 7.2 Performance Optimisation

No quantitative performance measurements are claimed for this mini-project. However, sound low-code design practices are incorporated to keep record retrieval responsive: using Auto Number identifiers and Lookup fields to avoid redundant text duplication,

filtering reports by status and date, and executing conflict validation via a single indexed criteria query rather than iterating over all records in a loop.

**Table 7.2:** Performance Optimisation Results

| Optimisation Applied | Component | Before | After | Improvement |
|---|---|---|---|---|
| Lookup relationships instead of duplicate text fields | Pets, Appointments, Medical Records Forms | Not measured | Not measured | Eliminates repetitive typing of owner, pet, and doctor details |
| Single criteria fetch for appointment conflict check | DS-03 (Appointments On Validate) | Not measured | Not measured | Checks matching Veterinarian + Date + Time directly without loops |
| Filtered report views by Status and Date | Appointment Schedule & History Reports | Not measured | Not measured | Simplifies locating Pending, Confirmed, or Completed bookings |

### 7.3 Security Features

Veterinary clinics handle sensitive client contact details (phone numbers, email addresses, residential addresses) and confidential clinical consultation records. Security and data privacy in the proposed system rely on: (1) authenticated access through Zoho Creator's supported sign-in mechanisms; (2) explicit configuration of role-based permissions so only authorized staff and veterinarians can view or modify records; (3) mandatory field and format validation on email, phone, and date inputs; (4) private application sharing settings (avoiding public links); and (5) strict use of fictional sample data in academic demonstrations.

**Table 7.3:** Security Controls Implemented

| Security Control | Where Applied | Threat Mitigated |
|---|---|---|
| Zoho Account Authentication | Application Sign-In | Prevents unauthenticated public access to clinic records |
| Configurable Role-Based Permissions | Pet Owners, Appointments & Medical Records | Restricts modification of clinical diagnoses and owner contact data (must be explicitly configured) |

| Security Control | Where Applied | Threat Mitigated |
|---|---|---|
| Mandatory Field & Format Validation | All 5 Forms + Deluge On Validate Rules | Reduces incomplete, malformed, or conflicting appointment records |
| Responsible Fictional Data Usage | Sample Records & Academic Report | Ensures no real client personal data or real clinical records are exposed |

### 7.4 Governance and Administration

Application governance involves maintaining consistent internal field link names across forms and Deluge scripts, verifying that lookup relationships reference Auto Number record IDs rather than display names, and periodically exporting backup copies of form records and application definitions.

Note that the availability of advanced governance features—such as multi-environment staging, audit trails, user role quotas, and daily automated email notification limits—depends on the specific Zoho Creator subscription plan and account configuration.

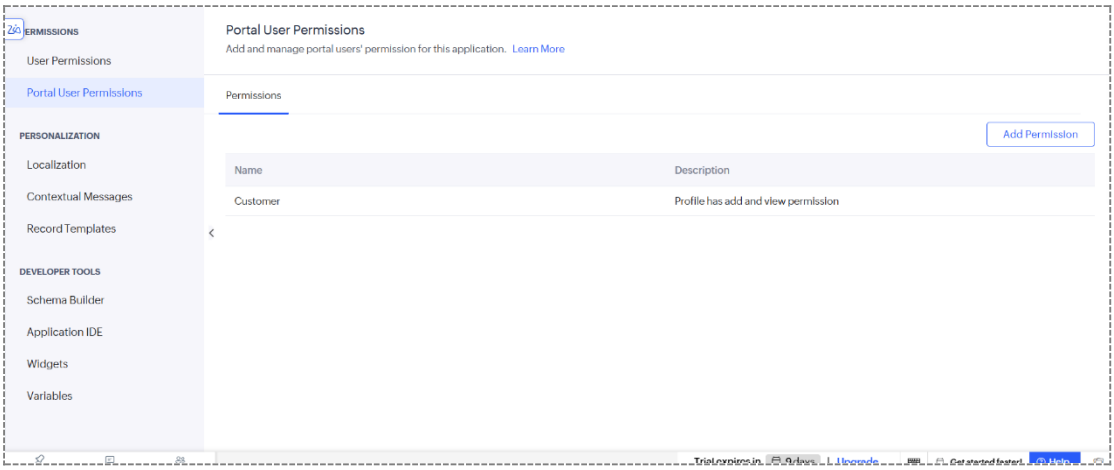



**Figure 7.2: Application Administration Settings (Placeholder)**

# CHAPTER 8
# RESULTS AND DISCUSSION

## 8.1 Application Output

The intended operational outcome of the Pet Care & Veterinary Appointment Management System is a cohesive low-code application that centralizes clinic administration across ten steps: registering a pet owner (e.g., fictional owner 'Ramesh Kumar'), registering a pet ('Bruno', Labrador, linked to Ramesh Kumar), maintaining veterinarian details ('Dr. Priya Nair', General Veterinary Care), creating a validated appointment booking, updating its status from Pending to Confirmed and Completed, recording the veterinarian's consultation diagnosis and prescription in Medical Records, and reviewing the updated schedules and Clinic Dashboard.

Figures 8.1 to 8.3 provide designated placeholder boxes for inserting screenshots of these operational steps once executed in the Zoho Creator workspace. No fabricated performance metrics or real-world clinic adoption claims are made.

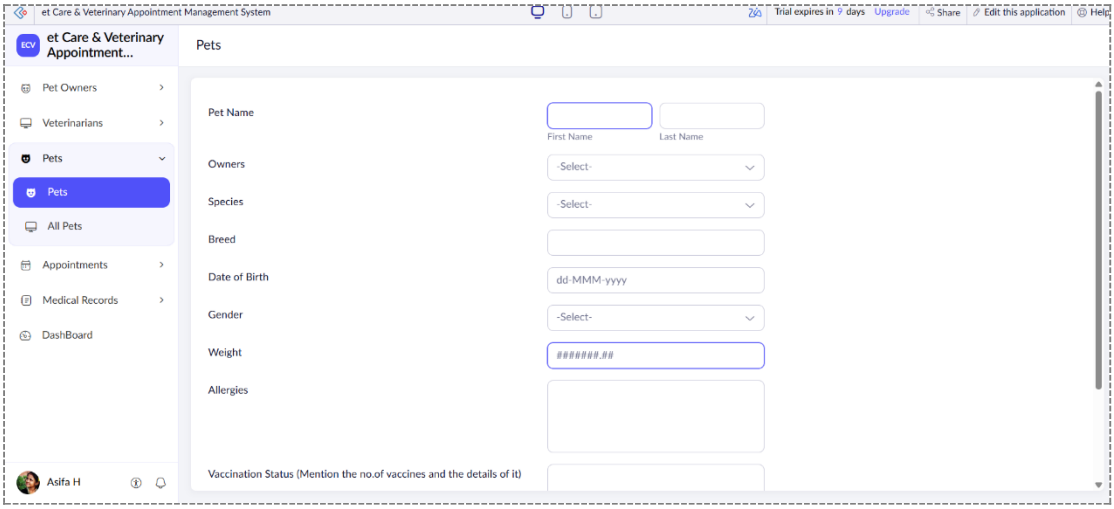



**Figure 8.1: Scenario Step 1 – Pet Owner and Pet Registration (Placeholder)**

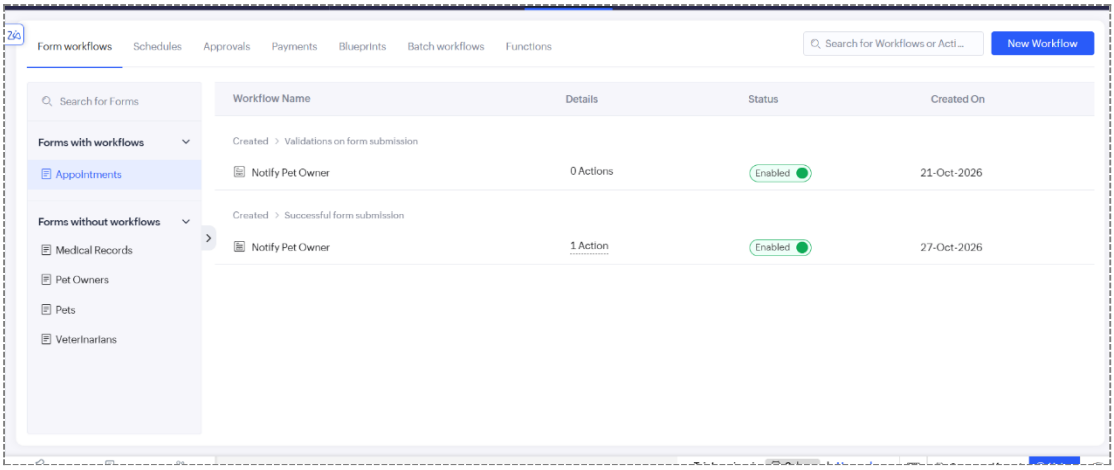



**Figure 8.2: Scenario Step 2 – Veterinary Appointment Scheduling (Placeholder)**

## 8.2 Achievement of Objectives

**Table 8.1:** Mapping of Objectives to Outcomes

| Objective (Sec. 1.3) | Implemented Through | Outcome / Evidence |
|---|---|---|
| Develop centralized pet owner & pet registration modules (Obj. 1–3) | Module 1 (Pet Owners Form) & Module 3 (Pets Form with Owner Lookup) | Structured schema and lookup linkage defined (Tables 4.2, 4.4) |
| Maintain veterinarian profiles, specializations & availability (Obj. 4) | Module 2 (Veterinarians Form) | Doctor ID, Specialization dropdown, Available Days, and Fee defined (Table 4.3) |
| Schedule and track veterinary appointments (Obj. 5, 7) | Module 4 (Appointments Form with Pet & Veterinarian Lookups) | Pending, Confirmed, Completed, and Cancelled workflow defined (Table 4.5) |
| Maintain consultation diagnosis, treatment & prescriptions (Obj. 6) | Module 5 (Medical Records Form with Appointment Lookup) | Clinical record structure defined for authorized veterinary entry (Table 4.5) |
| Create searchable reports and centralized dashboard (Obj. 8, 9) | Module 6 (4 Core Reports) & Module 7 (Dashboard Page) | All Pets, Schedule, History, Medical Reports & Dashboard designed (Table 5.1, Fig. 4.4) |
| Implement workflow automation and data validation (Obj. 10, 11) | Module 8 (Workflows A–D & Illustrative Deluge Scripts DS-01 to DS-04) | Status init, past-date guard, conflict check, and confirmation email designed (Sec. 5.5, 5.6) |
| Demonstrate low-code application development in veterinary care (Obj. 12) | Chapters 4, 5, and 8 | End-to-end low-code architecture and 12 validation test cases documented |

## 8.3 Comparison with the Existing System

**Table 8.2:** Existing System versus Proposed Application

| Parameter | Existing System | Proposed Application | Improvement |
|---|---|---|---|
| Information Management | Owner, pet, doctor, appointment, and medical records scattered across paper/files | Centralized in 5 interconnected Zoho Creator forms | Unified record management and easier retrieval |
| Data Entry Effort | Owner and pet details re-entered manually for every booking | Selected once via Lookup fields (Owner, Pet, Veterinarian) | Reduced repetitive administrative typing |
| Appointment Tracking & Conflicts | Manual diary checks prone to overlapping bookings | Structured status tracking (Pending/Confirmed/Completed/Cancelled) + Deluge conflict check | Improved schedule visibility and validation |
| Consultation History Retrieval | Manual search through physical consultation cards | Medical Records linked to Appointments and filterable via reports | Organized longitudinal pet medical histories |
| Application Prototyping & Modification | Requires full-stack coding for any schema change | Visual form/report builder and configurable Deluge workflows | Faster prototyping and easier form modification |

## 8.4 Limitations of the Low-Code Approach

Realistic technical and operational limitations of the proposed Pet Care & Veterinary Appointment Management System include:

- **Dependence on internet connectivity and platform plan limits: As a cloud-hosted Zoho Creator application, continuous internet access is required, and record capacity, workflow executions, and daily email notification quotas depend on the selected Zoho Creator plan.**
- **Need to configure and test scheduling conflict logic: Storing Available Days on the Veterinarians form or creating a basic Appointments form does not**

**automatically prevent double-booking; conflict prevention requires configuring and testing validation logic (Workflow C) against existing records.**

- **No autonomous medical diagnosis or treatment: The system is an administrative record-keeping tool; all diagnoses, treatments, and prescriptions must be entered by qualified veterinary personnel.**
- **Dependence on accurate data maintenance: Dashboard statistics and appointment schedules are only as reliable as the accuracy and timeliness of the records entered by clinic staff.**
- **Need for explicit permission and email configuration: Role-based access restrictions and automated confirmation emails (Workflow D) do not operate automatically unless explicitly configured and verified in the account.**
- **Exclusion of specialized clinical subsystems: Online payments, pharmacy inventory management, live GPS tracking, telemedicine, and external lab integrations are not included in the basic low-code model.**

### 8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

**No-code configuration: Creating the five forms (Pet Owners, Veterinarians, Pets, Appointments, Medical Records), configuring Auto Number and Lookup fields, setting dropdown choices, building tabular reports, and arranging Dashboard summary cards are accomplished visually using Zoho Creator's no-code builders.**

**Low-code scripting (Deluge): Enforcing business rules that span fields or existing records—such as rejecting past appointment dates (Workflow B), querying existing bookings to prevent veterinarian time-slot conflicts (Workflow C), and triggering conditional confirmation emails (Workflow D)—requires low-code Deluge scripting.**

**Pro-code boundaries: Full pro-code development would only be necessary if the clinic required custom real-time video telemedicine, hardware diagnostic device integration, or complex multi-branch ERP billing engines beyond standard Zoho Creator capabilities.**

# CHAPTER 9
# CONCLUSION AND FUTURE ENHANCEMENTS

## 9.1 Conclusion

The Pet Care & Veterinary Appointment Management System demonstrates how Zoho Creator's low-code development environment can be applied to solve administrative challenges in veterinary clinic management. By replacing fragmented paper registers with five interconnected forms—Pet Owners, Veterinarians, Pets, Appointments, and Medical Records—linked through Auto Number identifiers and Lookup fields, the proposed system streamlines owner and pet registration, structures appointment tracking across Pending, Confirmed, Completed, and Cancelled statuses, and organizes clinical consultation records.

Searchable reports and a centralized Dashboard Page provide clear visibility into clinic schedules, while form validation rules and illustrative Deluge workflows address status initialization, past-date rejection, appointment conflict checking, and confirmation alerts. While acknowledging platform dependencies and the boundary between implemented and proposed features, the project highlights the practical value of low-code engineering in rapidly prototyping structured administrative applications.

## 9.2 Future Enhancements

- Automated Pet Vaccination & Follow-Up Reminders: Configure scheduled workflows to send automated SMS or email reminders when a pet's Vaccination Status is 'Due Soon' or when a Medical Record's Follow-Up Date approaches.
- Advanced Appointment Calendar & Availability Management: Implement dynamic slot blocking based on veterinarian working hours, leave schedules, and consultation duration.
- Online Appointment Requests via Customer Portal: Configure a secure Zoho Creator Portal allowing registered pet owners to request appointments and view their pets' vaccination histories online.
- Billing, Invoicing, and Online Payment Management: Add automated invoice generation based on Veterinarian Consultation Fees and treatment charges, integrated with supported payment gateways.

- Veterinary Pharmacy, Medicine & Inventory Management: Link prescriptions in Medical Records to a clinic medicine inventory module to track stock levels and expiry dates.
- Enhanced Role-Based Access & Zoho Ecosystem Integration: Enforce granular field-level permissions for receptionists vs. veterinarians and integrate with supported Zoho communication/accounting tools.
- Clinic Utilization Analytics & Mobile Usability Enhancements: Expand dashboard analytics to track peak appointment hours, common consultation reasons, and mobile-optimized clinician views.

### 9.3 Learning Outcomes and Career Relevance

Designing and documenting the Pet Care & Veterinary Appointment Management System developed practical competencies in relational entity modeling, primary/foreign key design using Auto Number and Lookup fields, choice field standardization, tabular and dashboard reporting, event-driven Deluge scripting, and systematic test-case specification.

These skills directly support modern software engineering and low-code developer roles—including Low-Code Application Developer, Business Process Automation Specialist, and Solutions Analyst—by demonstrating how domain requirements can be translated into a clean, maintainable Zoho Creator architecture.

# REFERENCES


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# APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

**Table A.1:** Feature Coverage Checklist

| Unit | Zoho Creator Feature | Done | Where in the Application | Report Section | Figure / Code No. |
|---|---|---|---|---|---|
| I | Application created & sections/menu organised | Proposed / Designed | Pet Care & Vet System – 6 navigation sections | 4.5, 5.1 | Fig. 4.4, 5.1 |
| II | Forms with ≥ 8 distinct field types | Y (Designed) | Auto Number, Name, Single Line, Multi Line, Phone, Email, Address, Date, Drop Down, Multi Select, Decimal, Currency, Lookup | 4.3, 5.2 | Tables 4.2–4.5 |
| II | Lookup field (relationship between forms) | Y (Designed) | Pets.Owner, Appointments.Pet, Appointments.Veterinarian, Medical_Records.Appointment | 4.2, 5.2 | Fig. 4.2, Table 4.1 |
| II | Subform (parent–child data) | N | Normalized using separate forms linked via Lookup fields | 4.2 | Table 4.1 |
| II | Field validations / layout rules | Y (Designed) | Mandatory fields, Email/Phone format, Date & Conflict validation | 4.3, 5.5 | Tables 4.2–4.5, Code 5.1 |
| II | List report with filter, grouping, custom action | Y (Designed) | All Pets, Appointment Schedule, Appointment History, Medical Records Reports | 5.3 | Table 5.1 |
| II | Kanban / Calendar / Timeline / Map report | Optional / Proposed | Appointment Schedule Calendar view (where supported) | 5.3 | Table 5.1 |
| II | Pivot table / chart report | Proposed / Designed | Appointment Status Chart on Clinic Dashboard | 5.4, 5.9 | Fig. 4.4, 5.9 |
| II | Page / dashboard built with UI Builder | Proposed / Designed | Clinic Dashboard Page (Total Pets, Today's/Pending/Completed/Upcoming Appts) | 4.5, 5.4 | Fig. 4.4, 5.5 |

| Unit | Zoho Creator Feature | Done | Where in the Application | Report Section | Figure / Code No. |
|---|---|---|---|---|---|
| II | Deluge – form events (validate / success / field actions) | Proposed (Illustrative) | DS-01 (On Load), DS-02 & DS-03 (On Validate), DS-04 (On Success) | 5.5 | Code 5.1, 5.2 |
| II | Deluge – functions, fetch, loops, collections | Proposed (Illustrative) | Criteria fetch and .count() in DS-03 conflict check; lookup fetch in DS-04 | 5.5 | Code 5.1, 5.2 |
| II | Email / in-app notifications | Proposed (If Configured) | Workflow D: Appointment confirmation email to pet owner | 5.5, 5.6 | Code 5.2, Table 5.3 |
| II | Form workflow | Proposed / Designed | Workflows A, B, C, and D on Appointments form | 5.6 | Table 5.3 |
| II | Scheduled workflow | N (Future Enhancement) | Proposed for future vaccination & follow-up reminders | 9.2 | Sec. 9.2 |
| II | Approval process | N | Not required for clinic appointment scheduling | 5.6 | — |
| II | Blueprint | N | Status tracked via Appointment Status dropdown | 5.2, 5.6 | Table 4.5 |
| II | Batch / report workflow | N | Not used in current scope | 5.6 | — |
| II | Testing & debugging evidence | Designed (Pending Test) | 12 formal test scenarios (TC01–TC12) specified; execution pending | 6.1–6.4 | Table 6.1, 6.2 |
| III | Integration with a Zoho service | Optional (Built-in Mail) | Built-in sendmail action in Workflow D (if configured) | 5.7 | Table 5.4 |
| III | Connection / REST API call (invokeurl) | N | No external REST API used | 5.7 | Code 5.3 |
| III | Webhook or Creator REST API | N | Not used in current academic mini-project | 5.7 | — |
| III | Users, roles and permission profiles | Designed (Configurable) | Role–Permission Matrix for Admin, Clinic Staff, and Veterinarian | 4.6, 5.8 | Table 4.6 |

| Unit | Zoho Creator Feature | Done | Where in the Application | Report Section | Figure / Code No. |
|---|---|---|---|---|---|
| III | Record-level access criteria / customer portal | N (Future Enhancement) | Public customer portal is out of scope (future enhancement) | 1.4, 9.2 | Sec. 9.2 |
| III | Zia / AI feature | N | No AI diagnosis used; medical records entered by veterinarians | 1.4, 5.9 | Sec. 5.9 |
| III | Analytics (charts / Zoho Analytics) | Proposed / Designed | Dashboard summary metrics and Appointment Status Chart | 5.4, 5.9 | Fig. 4.4 |
| IV | Performance optimisation applied & measured | Design Practices Only | Lookup normalization and single-query conflict check (not measured) | 7.2 | Table 7.2 |
| IV | Deployment across environments | Academic Workspace | Developed in Zoho Creator workspace (not claimed as commercial deployment) | 7.1 | Table 7.1 |
| IV | Security controls (encryption, OAuth, permissions) | Designed | Authentication, validation, configurable permissions, fictional data | 7.3 | Table 7.3 |
| IV | Governance (audit trail, administration) | Designed | Auto Number IDs, consistent link names, plan limit awareness | 7.4 | Sec. 7.4 |
| V | Pro-code extension (Catalyst / widgets / custom functions) | N | Not used (justified in Section 5.10) | 5.10 | Sec. 5.10 |
| V | Low-code limitations & spectrum analysis | Y | Limitations and No-Code/Low-Code/Pro-Code spectrum analyzed | 8.4, 8.5 | Sec. 8.4, 8.5 |
| — | Mobile app / portal access | N (Future Enhancement) | Web browser access; mobile/portal listed under future enhancements | 5.11, 9.2 | Sec. 5.11 |

**Total features documented in system specification: 15 designed/proposed core low-code features / 32 (Live execution verification pending)**

# APPENDIX B – COURSE OUTCOME MAPPING

**Table B.1:** Mapping of Course Outcomes to Report Chapters

| CO | Course Outcome Statement | Evidenced In | Self-Assessed Level | Examiner Level |
|---|---|---|---|---|
| **CO1** | Explain the principles, architecture, and business value of low-code application development using Zoho Creator. | Ch. 1, 2 | 3 | |
| **CO2** | Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting. | Ch. 4, 5 (5.1–5.6), 6 | 3 | |
| **CO3** | Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security. | Ch. 4.6, 5.7–5.9 | 2 | |
| **CO4** | Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices. | Ch. 7 | 2 | |
| **CO5** | Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer. | Ch. 5.9, 5.10, 8.4, 8.5, 9.3 | 3 | |

# APPENDIX C – COMPLETE DELUGE SOURCE CODE

### *C.1 Appointments Form – Illustrative Deluge Workflows (DS-01 to DS-04)*

```
//
================================================================================
// PET CARE & VETERINARY APPOINTMENT MANAGEMENT SYSTEM – ILLUSTRATIVE DELUGE
CODE
// Note: Verify exact internal field link names in Zoho Creator Form Builder
//
================================================================================

// --- 1. WORKFLOW A (DS-01): Initialize Appointment Status ---
// Trigger: Appointments Form -> Created -> On Load
if(input.Status == null || input.Status == "")
{
    input.Status = "Pending";
}

// --- 2. WORKFLOW B (DS-02): Appointment Date Validation ---
// Trigger: Appointments Form -> Created or Edited -> On Validate
if(input.Appointment_Date == null || input.Appointment_Date <
zoho.currentdate)
{
    alert "Validation Error: Appointment Date cannot be in the past.";
    cancel submit;
}

// --- 3. WORKFLOW C (DS-03): Veterinarian Appointment Conflict Check ---
// Trigger: Appointments Form -> Created or Edited -> On Validate
conflicting_records = Appointments[Veterinarian == input.Veterinarian &&
Appointment_Date == input.Appointment_Date && Appointment_Time ==
input.Appointment_Time && Status != "Cancelled" && ID != input.ID];
if(conflicting_records.count() > 0)
{
    alert "Scheduling Conflict: The selected veterinarian already has an
active appointment at this date and time slot.";
    cancel submit;
}

// --- 4. WORKFLOW D (DS-04): Appointment Confirmation Email Notification ---
// Trigger: Appointments Form -> Edited -> On Success (When Status changes to
'Confirmed')
// Note: Executes only if email notification functionality is configured in
the account
if(input.Status == "Confirmed")
{
    pet_record = Pets[ID == input.Pet];
    owner_record = Pet_Owners[ID == pet_record.Owner];
    if(owner_record.Email != null && owner_record.Email != "")
    {
        sendmail
        [
```

```
            from : zoho.adminuserid
            to : owner_record.Email
            subject : "Appointment Confirmed for " + pet_record.Pet_Name
            message : "Dear " + owner_record.Owner_Name + ", your veterinary
appointment for " + pet_record.Pet_Name + " on " + input.Appointment_Date + "
at " + input.Appointment_Time + " is confirmed."
        ];
    }
}
```

# APPENDIX D – USER MANUAL

## D.1 Accessing the Application

Open a supported web browser, sign in to Zoho Creator using the authorized clinic account, and select the 'Pet Care & Veterinary Appointment Management System' application. The centralized Dashboard Page is displayed first, and the five form modules and four reports are accessible from the navigation menu.

## D.2 Instructions for Clinic Staff and Veterinarians

- 1. Register Pet Owner: Open the Pet Owners form, enter Owner Name, Phone, Email, Address, and Registration Date, and click Submit.
- 2. Maintain Veterinarian Profiles: Open the Veterinarians form, enter Doctor Name, select Specialization (General Veterinary Care, Dermatology, Surgery, Dentistry), Phone, Email, Available Days, and Consultation Fee, and click Submit.
- 3. Register Pet Profile: Open the Pets form, enter Pet Name, select the registered Owner from the Lookup dropdown, choose Species, Breed, Date of Birth, Gender, Weight, Allergies, and Vaccination Status, and submit.
- 4. Schedule Appointment: Open the Appointments form, select the Pet and Veterinarian via Lookup fields, choose the Appointment Date, Time, and Reason, and submit (Status initializes to Pending).
- 5. Handle Validation Alerts & Update Status: If a date or scheduling conflict alert appears, select a valid future date/time slot. Update the appointment Status to Confirmed, Completed, or Cancelled as the visit progresses.
- 6. Record Medical Consultation: After examination, authorized veterinary staff open the Medical Records form, select the Appointment via Lookup, and enter Diagnosis, Treatment, Prescription, Follow-Up Date, and Veterinarian Notes.
- 7. Search Reports & View Dashboard: Use the All Pets, Appointment Schedule, Appointment History, and Medical Records reports to filter records, and open the Dashboard to review clinic summary counts.

### D.3 Instructions for the Application Administrator

- 1. Sign in with the application owner account and open the Zoho Creator builder to manage forms, choice fields, lookup links, reports, workflows, and the Dashboard Page.
- 2. Verify internal field link names before enabling Deluge scripts, and execute all twelve test cases in Table 6.1 (TC01 to TC12) using fictional sample data.
- 3. Configure role permissions explicitly and avoid publishing the application publicly so that pet owner contact details and clinical records remain protected.

# APPENDIX E – EVALUATION RUBRIC

*(For Examiner Use Only)*

| **Student Name** | ASIFA H | Register No. | 241501025 |
|---|---|---|---|
| **Project Title** | PET CARE & VETERINARY APPOINTMENT MANAGEMENT SYSTEM | Date | |

**Table E.1:** Evaluation Rubric for the Individual Project

| S.No. | Criterion | Indicators | Max. Marks | Marks Awarded |
|---|---|---|---|---|
| 1 | Problem Definition & Requirement Analysis | Clarity of problem, measurable objectives, complete FR/NFR, use cases | 10 | |
| 2 | System Design | Architecture, ER/data model, form specifications, workflow design, permission matrix | 10 | |
| 3 | Forms, Reports & Pages | Variety of field types, lookups/subforms, report types, UI Builder dashboards | 15 | |
| 4 | Deluge Scripting & Workflows | Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows | 15 | |
| 5 | Integration, RBAC & AI | Zoho/external integrations, connections, roles & permissions, Zia/analytics | 10 | |
| 6 | Testing, Deployment, Security & Governance | Test coverage, environments, optimisation, security controls, audit | 10 | |
| 7 | Live Demonstration | End-to-end working of the application across roles; feature checklist verified | 15 | |
| 8 | Project Report | Adherence to template and format, clarity, figures, references | 5 | |
| 9 | Viva-Voce | Understanding of the platform, design decisions and low-code concepts | 10 | |
| | **TOTAL** | | **100** | |

**Table E.2:** Performance Level Descriptors

| Level | % of Max. Marks | Descriptor |
|---|---|---|
| **Excellent** | 90–100 | Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently. |
| **Good** | 70–89 | Feature implemented correctly with minor gaps; explanation adequate. |
| **Satisfactory** | 50–69 | Feature partially implemented or works only for the main path; limited explanation. |
| **Needs Improvement** | < 50 | Feature missing, non-functional or not understood by the student. |

**Signature of Internal Examiner**

Name: _____

**Signature of External Examiner**

Name: _____

# APPENDIX F – VIVA-VOCE RECORD SHEET

*(For Examiner Use Only)*

| | | | |
|---|---|---|---|
| **Student Name** | ASIFA H | Register No. | 241501025 |
| **Department / Section** | Artificial Intelligence and Machine Learning | Date of Viva | |
| **Application URL Verified** | Yes / No | Demo Duration | _____ min |

**Table F.1:** Viva-Voce Questions and Responses

| Q.No. | Question / Topic Asked | Unit | Response (E / G / S / NI) | Remarks |
|---|---|---|---|---|
| 1 | | | | |
| 2 | | | | |
| 3 | | | | |
| 4 | | | | |
| 5 | | | | |
| 6 | | | | |
| 7 | | | | |
| 8 | | | | |

*E – Excellent  G – Good  S – Satisfactory  NI – Needs Improvement*

**Table F.2:** Marks Summary

| Component | Max. Marks | Marks Awarded |
|---|---|---|
| Criteria 1–8 (Appendix E) | 90 | |
| Viva-Voce | 10 | |
| **TOTAL** | **100** | |

**Examiners' Remarks:**

_____

_____

**INTERNAL EXAMINER** **EXTERNAL EXAMINER**

Name: _____ Name: _____